

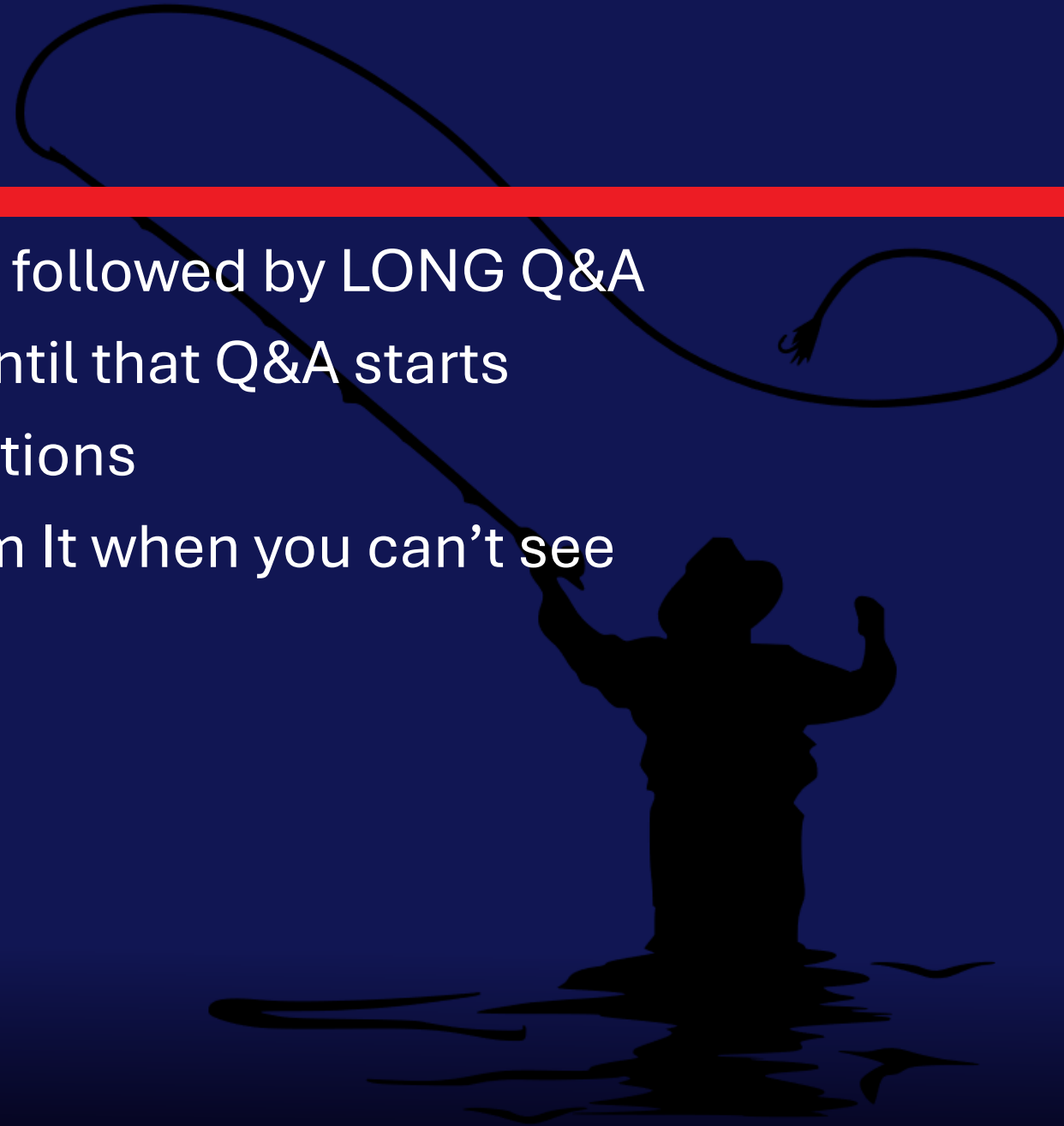
Ensuring Business Continuity

Challenges in Modern Endpoint Management



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



Carl Johan Hederoth

Product and Software Engineering
2Pint Software



Mike Terrill

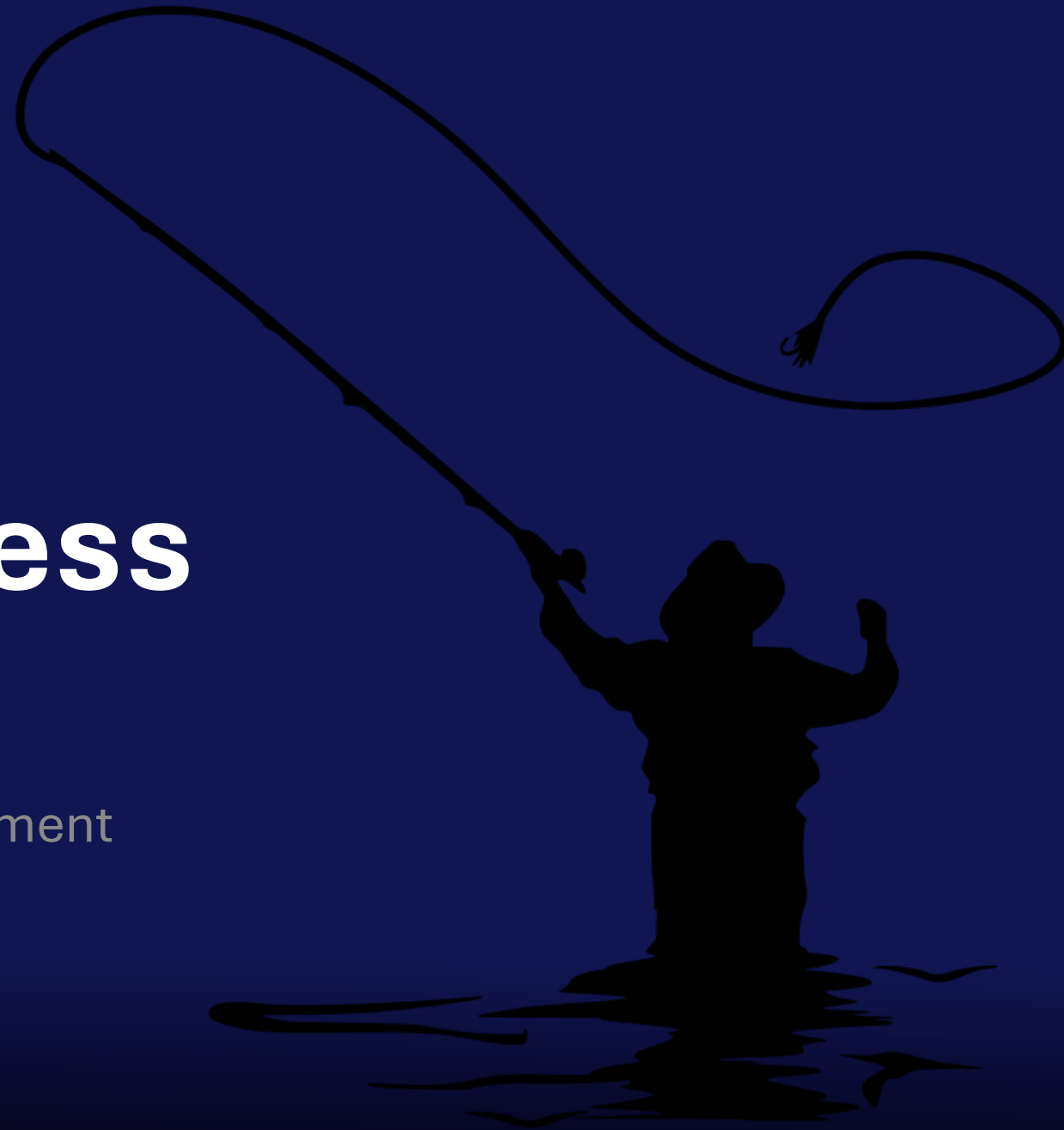
Principal Engineer, Microsoft MVP
2Pint Software

 @miketerrill

Can't see our slides? Can't hear? Need to repeat the question? Call us out!

Ensuring Business Continuity:

Challenges in Modern Endpoint Management



DISASTER RECOVERY

Jaguar Land Rover Operations 'Severely Disrupted' by Cyberattack

The automotive company said it disconnected its systems, which severely impacted both retail and manufacturing operations.

TECH MAR 23 2016, 5:16 PM ET

Three U.S. Hospitals Hit in String of Ransomware Attacks

by CONNOR MANNION

6 AM ET)



Cyber attack on M&S leads to click and collect delays

2 days ago

Share Save

Liv McMahon & Chris Vallance
Technology reporters



House of Commons hit by cyberattack from threat actor': internal email

Breached data includes information about government employees, devices



Kate McKenna · CBC News · Posted: Aug 14, 2025 4:00 AM EDT | Last Updated: August 14

City of Oakland Targeted by Ransomware Attack, Work Continues to Secure and Restore Services Safely



44 NHS appointments disrupted by criminal attack

Two hospitals, p... by ransomware attack in a month

'Payload Sent Two Months Before Attack, Backup Was Targeted Too'

WHAT EMERGES FROM LATEST ATTACKS

SYSTEMS VULNERABLE, ALERTS NOT TIMELY
A few of the recent attacks have been following the Phobos template of ransomware, which is around since 2018 but has evolved with successful decryption attempts

BE CYBERSAFE

HELPLINE: Dial 1930, the nationwide phone number, to report cybercrimes and seek assistance from authorities
PROFILE LOCKING: Use profile locking features on



UK Retailer Co-op Confirms Data Breach Affecting 6.5 Million Members

GAS SHORTAGES PERSIST IN WAKE OF COLONIAL PIPELINE CYBERATTACK

Gov. Larry Hogan | (R) Maryland



9:38 AM ET

ING EXCLUSIVE CNN INVESTIGATION ► CNN TEAM WITNESSED ERITREAN STATE OF THE UNION

Ransomware Attack Strikes U.S. Meat Operations at JBS

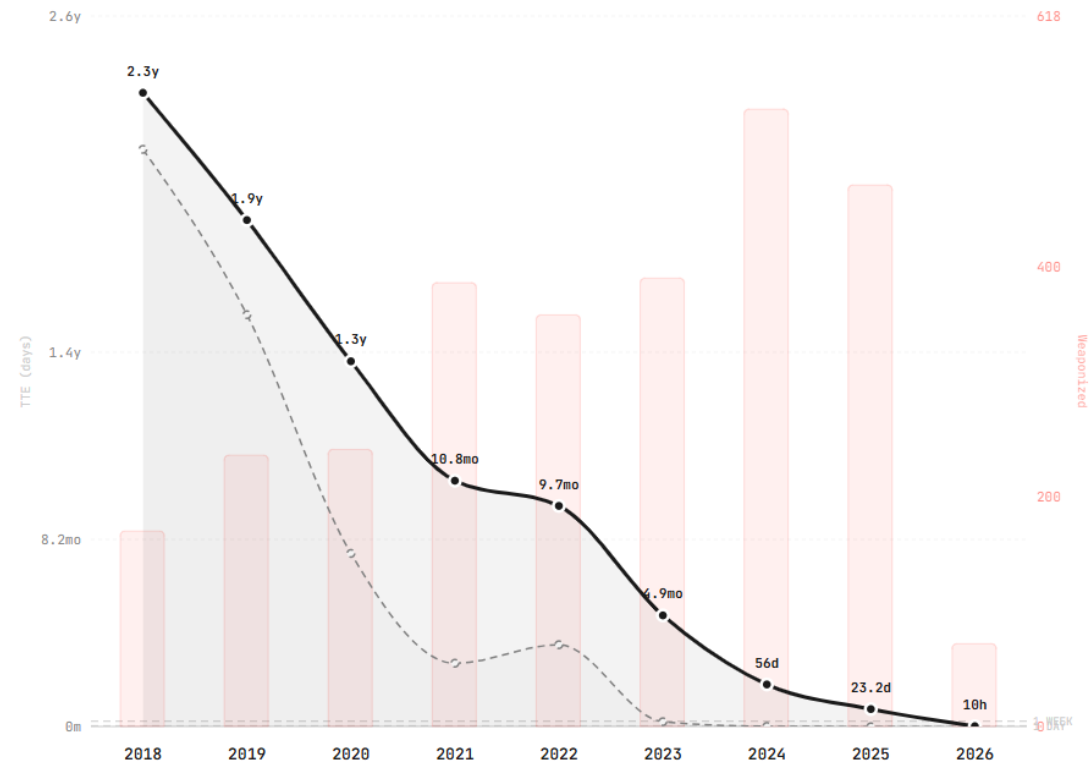


Zero Day Clock

From Vulnerability to Exploitation

TTE (Time-to-Exploit) measures the gap between CVE disclosure and confirmed exploitation

— Mean TTE (10% trimmed, days) --- Median TTE (days) ■ Weaponized Exploits (count)



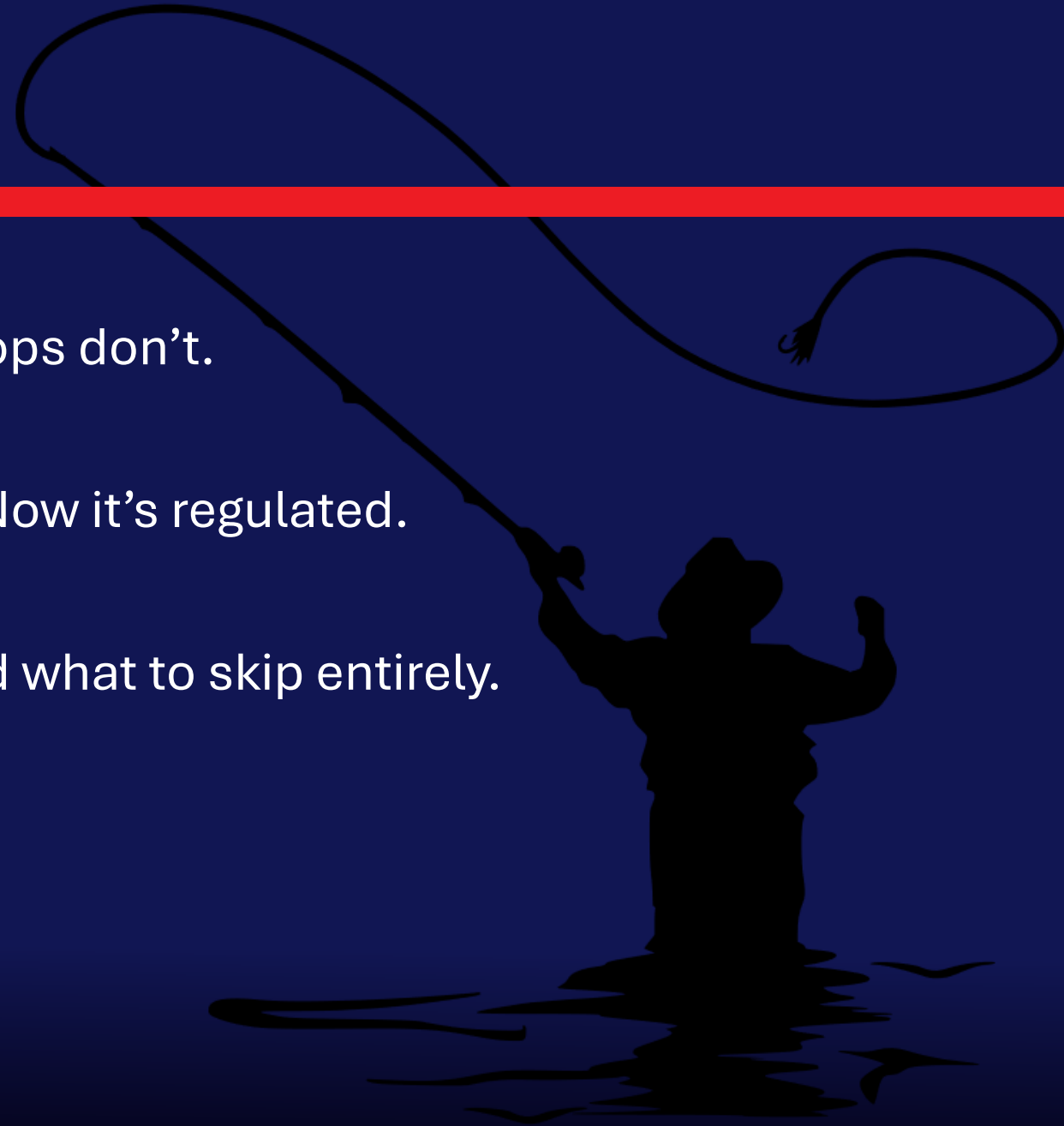
Based on 3,532 CVE-exploit pairs from trusted sources (CISA KEV, VulnCheck KEV & XDB)

• zerodayclock.com



Overview

- ◆ The Problem
 - ◆ Your servers have a plan. Your laptops don't.
- ◆ Why now
 - ◆ Downtime was always expensive. Now it's regulated.
- ◆ What works
 - ◆ The right tool for right scenario. And what to skip entirely.



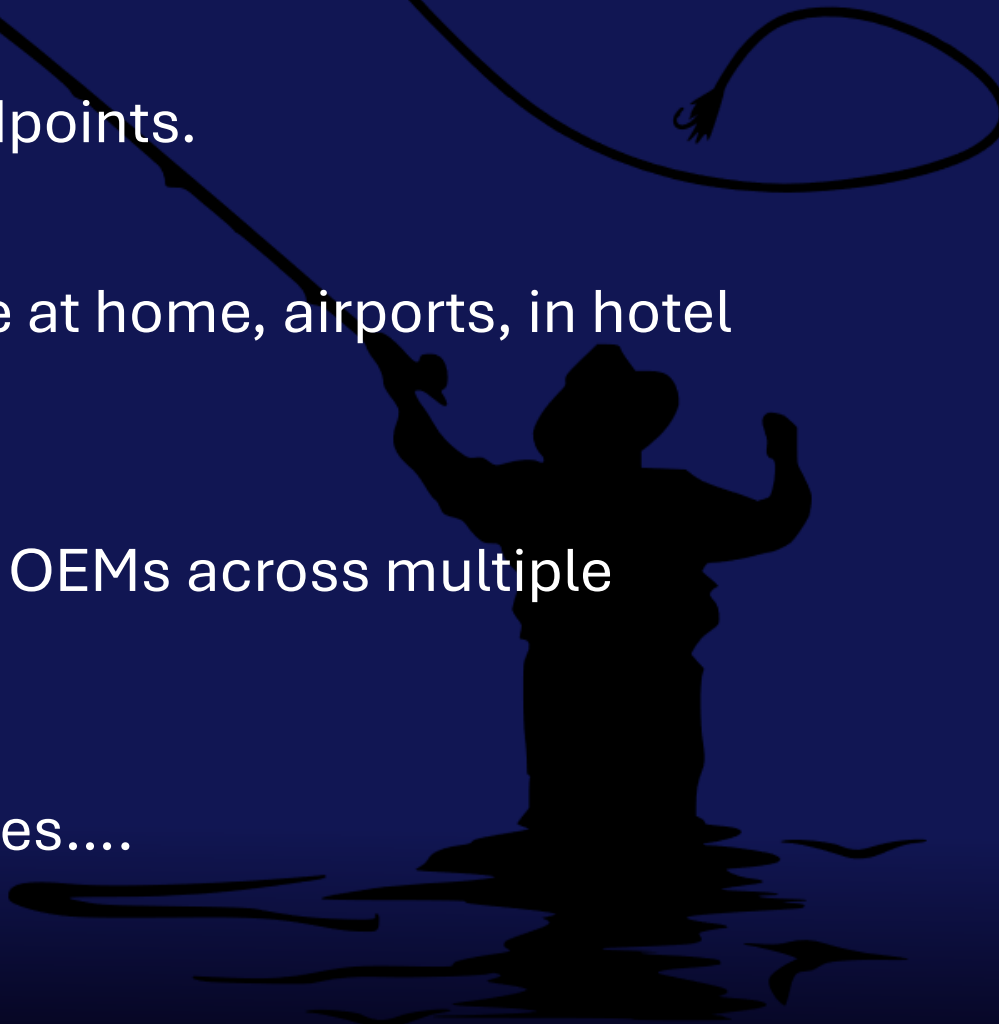
Do you have a recovery strategy?

- ◆ Who has recovery plan that covers endpoint?
- ◆ Who has tested it in the last 12 month?
- ◆ Could rebuild 10% of you fleet by Friday? 25%, 50%?

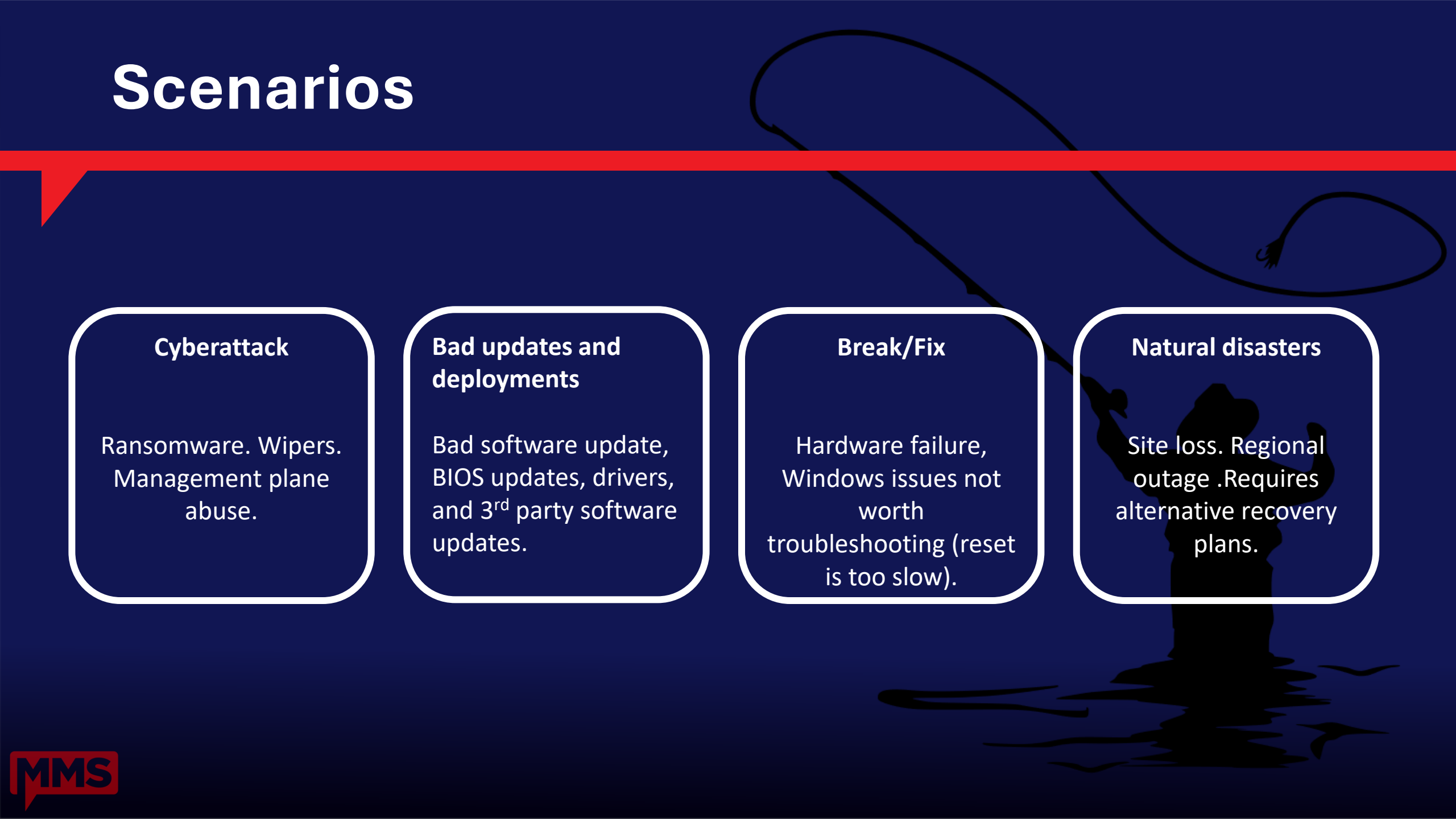


Why endpoints are different

- ◆ Scale
 - ◆ Hundreds of servers. Tens of thousands endpoints.
- ◆ Distribution
 - ◆ Servers are in your data center. Endpoint are at home, airports, in hotel rooms.
- ◆ Heterogeneity
 - ◆ A few server SKU. Endpoints might have five OEMs across multiple generations.
- ◆ User
 - ◆ Profiles, compliance, support desk, deadlines....



Scenarios



Cyberattack

Ransomware. Wipers.
Management plane
abuse.

Bad updates and deployments

Bad software update,
BIOS updates, drivers,
and 3rd party software
updates.

Break/Fix

Hardware failure,
Windows issues not
worth
troubleshooting (reset
is too slow).

Natural disasters

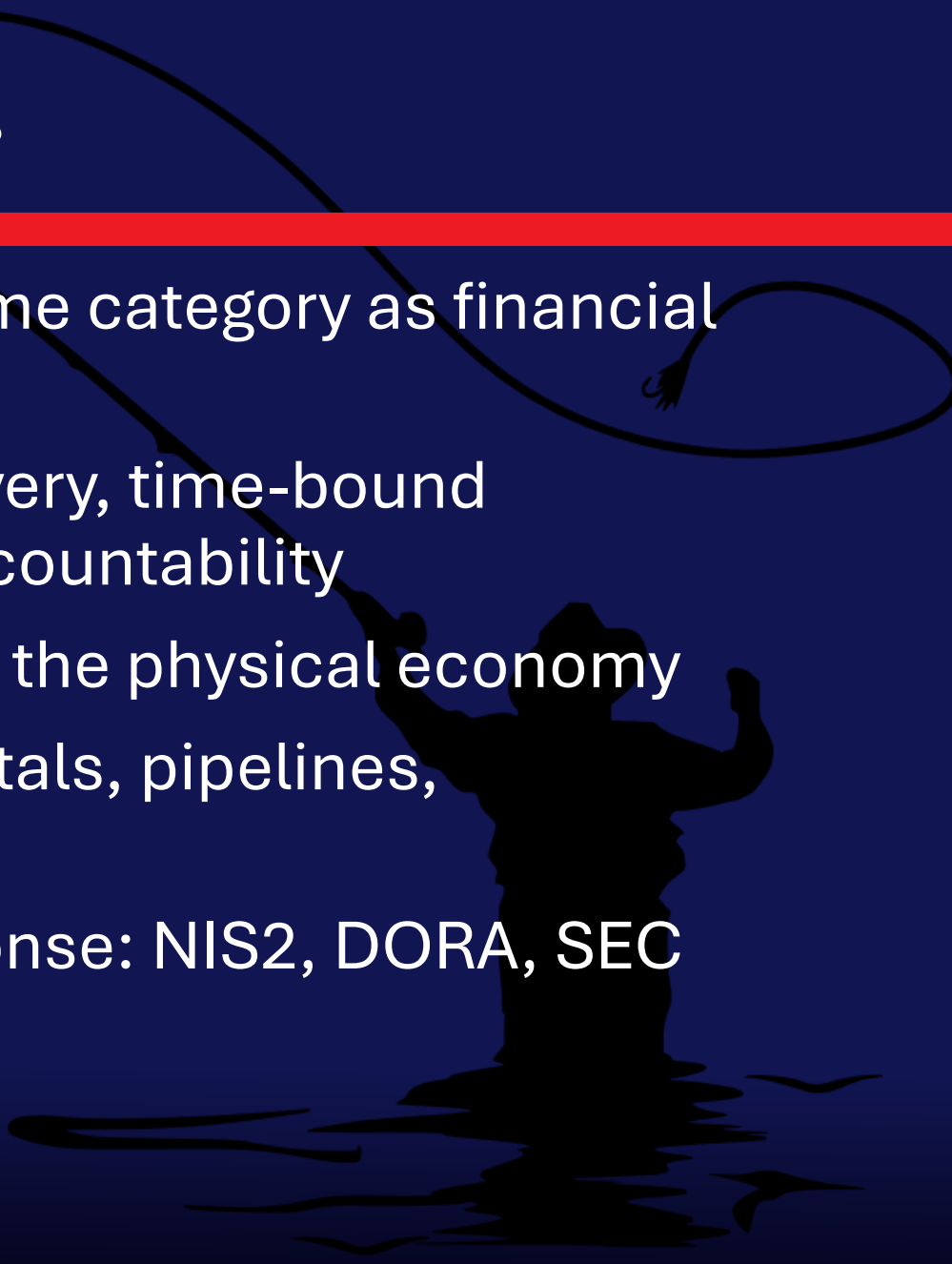
Site loss. Regional
outage .Requires
alternative recovery
plans.

The business impact

Industry	Typical Impact	Recent Example
Retail	~\$10M annual median downtime; \$13k-\$130K/hour POS outage	CrowdStrike crash (2024); Fortune 500 retail losses \$5.4B
Healthcare	\$7.4M avg cost per breach; lowest ransom median ~\$150k	WannaCry hit the NHS. ~19000 appointments cancelled. Cost est.: £92M+
Airlines/Travel	\$500–\$550M per major outage	Delta CrowdStrike (2024): 7,000+ flights cancelled, 1.3M passengers stranded
Public Services	Median ransom \$2.5M; recovery costs and rising	Aug 2025: 200 Swedish municipalities hit by ransomware
Construction/ Industrial	\$25k–\$500k/hr downtime; 70% of industrial ransomware in manufacturing	Norsk Hydro – Ransomware spread to 22K endpoints/170 sites, forcing manual operations in smelting plants; estimated cost \$70M+

The regulators caught up!

- ♦ Cyber is now operational resilience – same category as financial controls and physical safety
- ♦ **What they all now require:** tested recovery, time-bound restoration, fast disclosure, personal accountability
- ♦ 2017 – NotPetya proved cyber could halt the physical economy
- ♦ 2020-2022 – ransomware wave hit hospitals, pipelines, governments
- ♦ 2023-2025 – coordinate regulatory response: NIS2, DORA, SEC Cyber, NIST CSF 2.0
- ♦ 2025-2026 – First enforcement actions



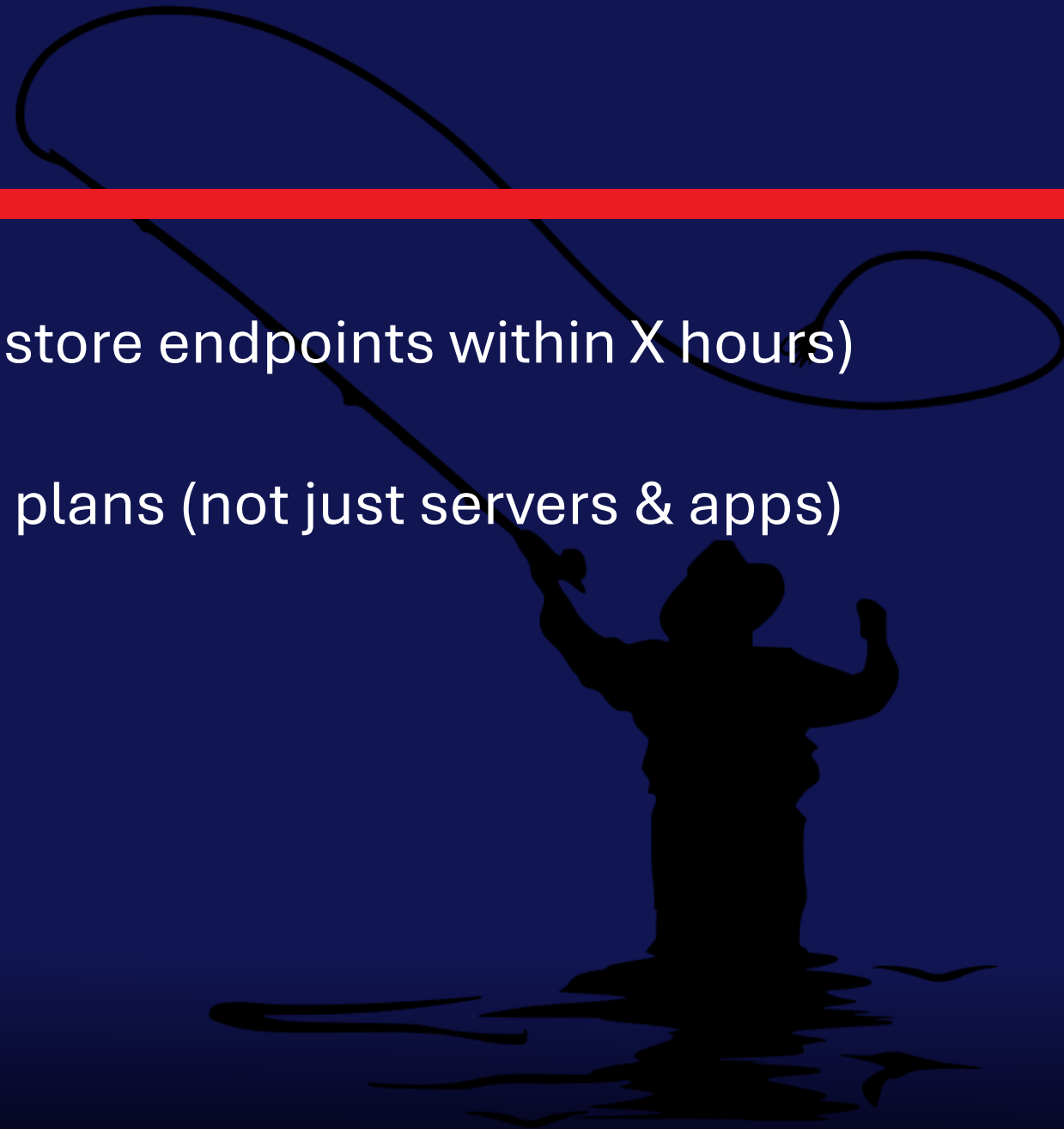
Different jurisdiction. Same expectations.

Framework	Region/Scope	What it means to endpoint recovery
NIST CSF 2.0	Global (US-origin)	“Recover” is not a top-level function – explicit expectation to restore assets and operation
DORA	EU – Financial Services	Annual testing of ICT recovery plans, including cyberattack scenarios. Endpoints in scope.
NIS2	EU – Critical Sectors	Recovery procedures must restore systems withing acceptable timeframes. Periodic testing required.
SEC Cyber Rule	US – Public companies	Material incidents disclosed withing 4 business days. Slow recovery = public bad news
HIPAA Security	US – Healthcase	Contingency plan standard: backup, DR, emergency mode operation – all required

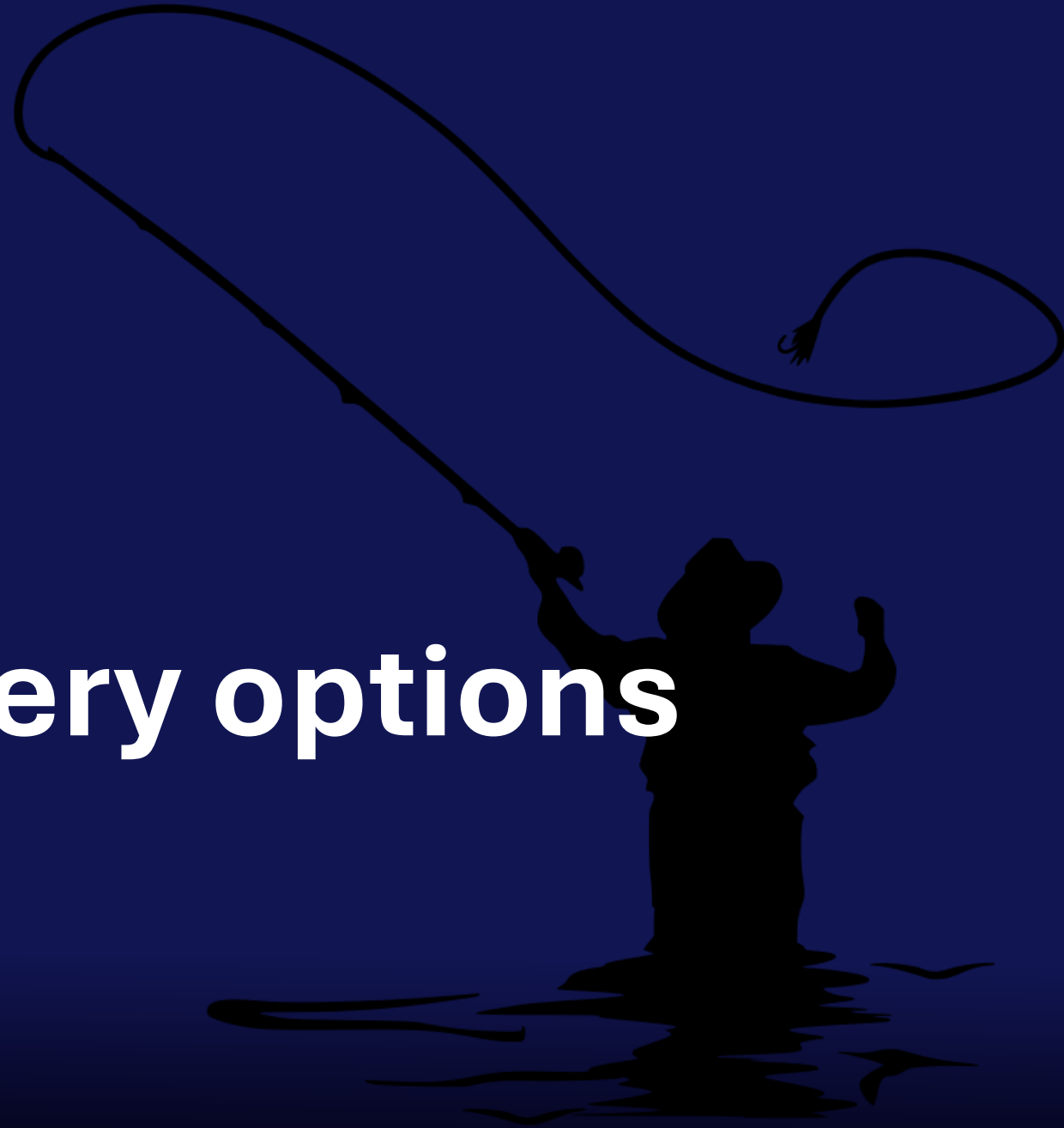
Non-compliance isn't cheap: NIS2 / DORA up to €10M or 2% of global turnover. HIPAA up to \$2.1M per violation category/year. SEC enforcement actions \$1M–\$7M. **Plus personal liability for leadership under NIS2 and DORA.**

Where to start

1. Incident response
Maintain recovery SLAs (e.g., restore endpoints within X hours)
2. Business continuity
Integrate endpoints into DR/BC plans (not just servers & apps)
3. Recovery assurance
Automate endpoint recovery

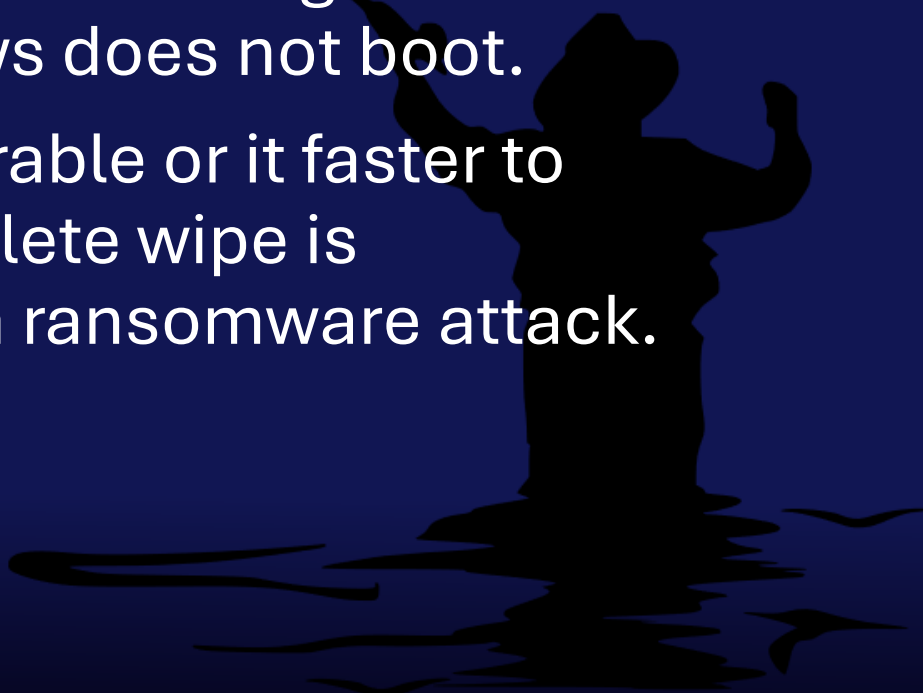


Endpoint recovery options



Endpoint recovery options

1. **Alternate OS** – user needs to get work done now!
Let the user connect to remote PC, while IT is working on plan b.
2. **Repair Windows** – maybe there is an easy fix that can be automated to repair systems in bulk. PXE booting and task sequences work well for when Windows does not boot.
3. **Reinstall Windows** – OS is not recoverable or it faster to reinstall Windows from scratch. Complete wipe is recommended when recovering from a ransomware attack.



Alternate OS: Boot to W365 or VDI w/(i)PXE

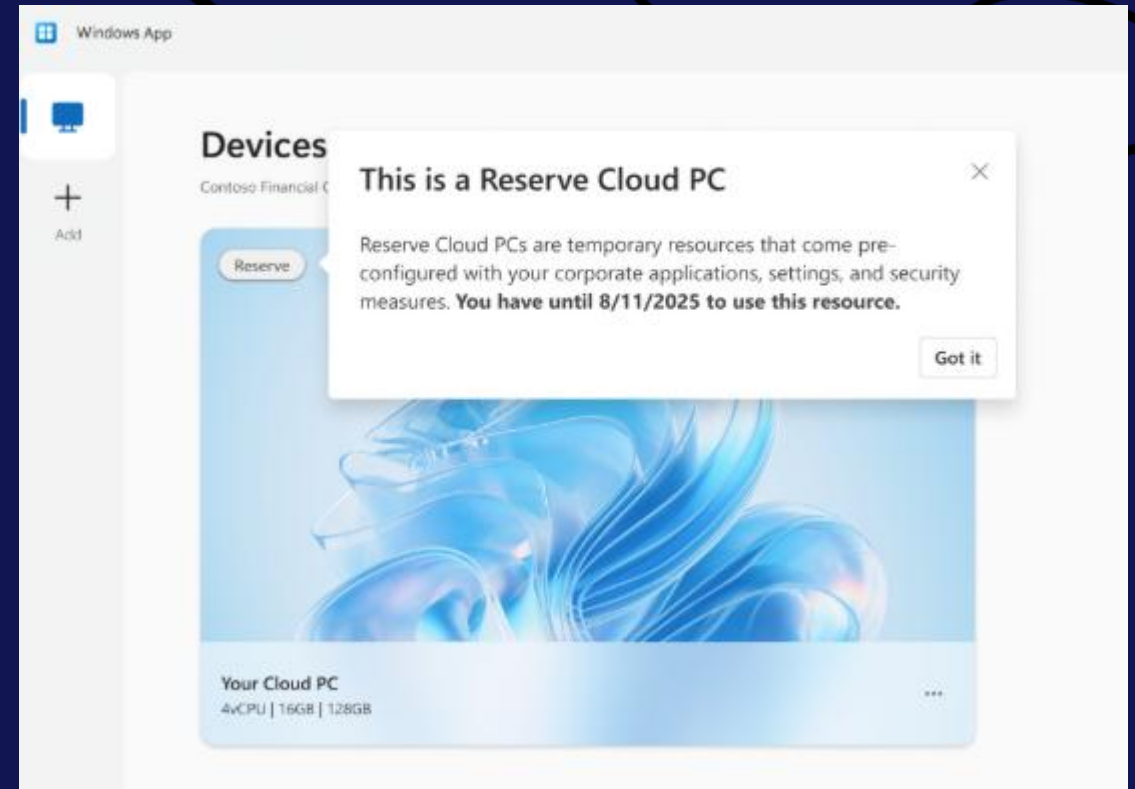
- ◆ OS and or hard drive may not be able to boot
- ◆ WinRE not working (or non-existent)
- ◆ Otherwise, device is still functional
- ◆ End user needs to work now!
- ◆ (i)PXE boot device to alternate OS
- ◆ Connect to W365 or VDI



Alternate OS: Windows 365 Reserve

Windows 365 Reserve is a new offering that provides:

- ◆ **Temporary Cloud PC access** for quick and secure productivity when a user's physical device is unavailable or short-term access is needed.
- ◆ **Pre-configured environment** with Microsoft 365 apps, corporate settings, and security policies, centrally managed via Microsoft Intune.
- ◆ **Flexible connectivity** — users can securely connect from any device using the Windows App or web browser for a consistent Windows experience.



Repair Windows: Quick machine recovery

← More recovery options



Quick machine recovery

Let your device search for solutions to repair itself



Troubleshoot

Reset your PC or see advanced options

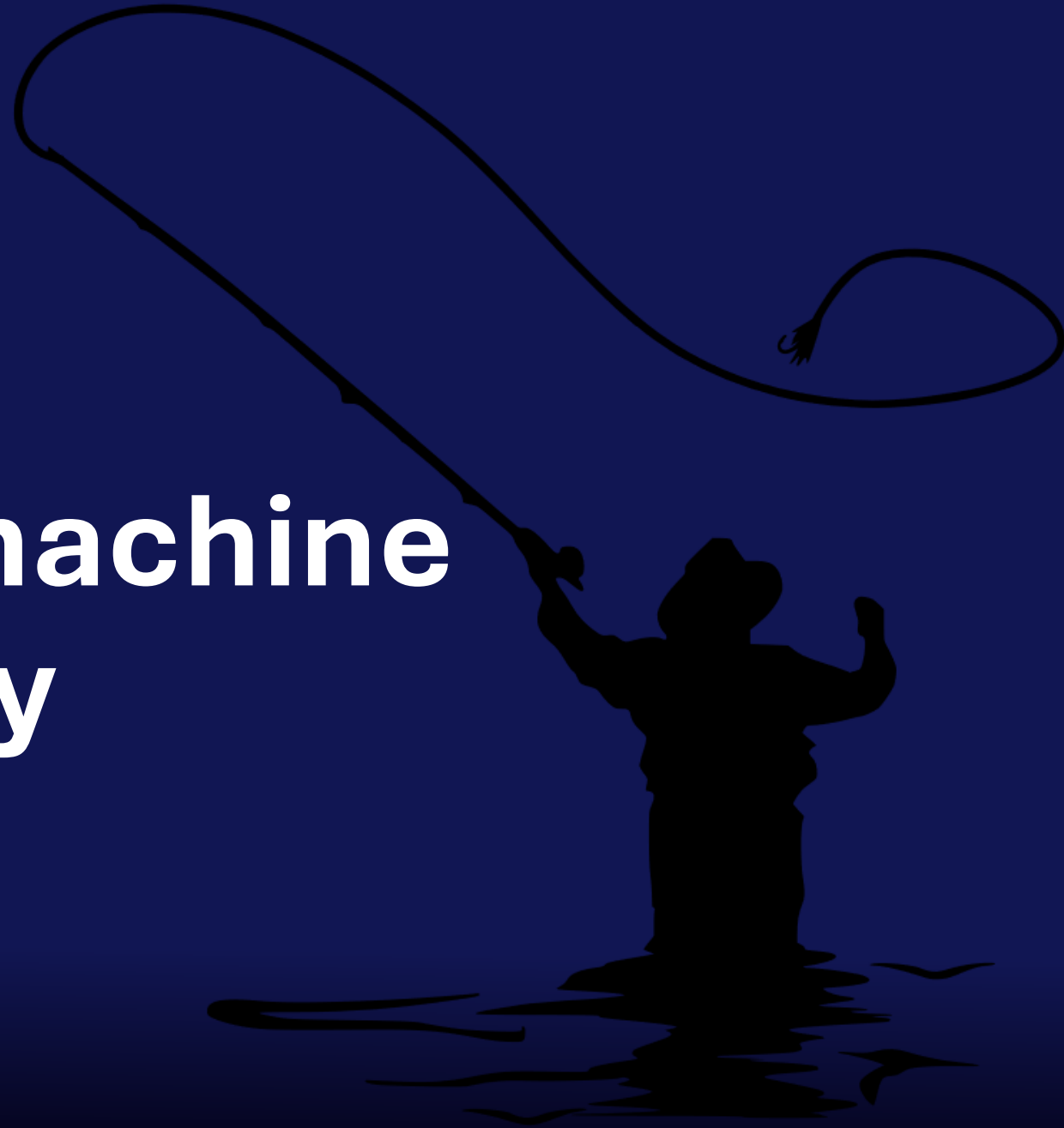


Turn off your PC

- ◆ A feature that enables recovery when Windows devices cannot boot
- ◆ Automatically search for remediations
- ◆ Uses the Windows Recovery environment to scan Windows Update

DEMO

Quick machine recovery



About Windows



Windows 11

Microsoft Windows

Version 24H2 (OS Build 26120.3941)

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Administrator
2Pint Software

OK



87°F
Sunny

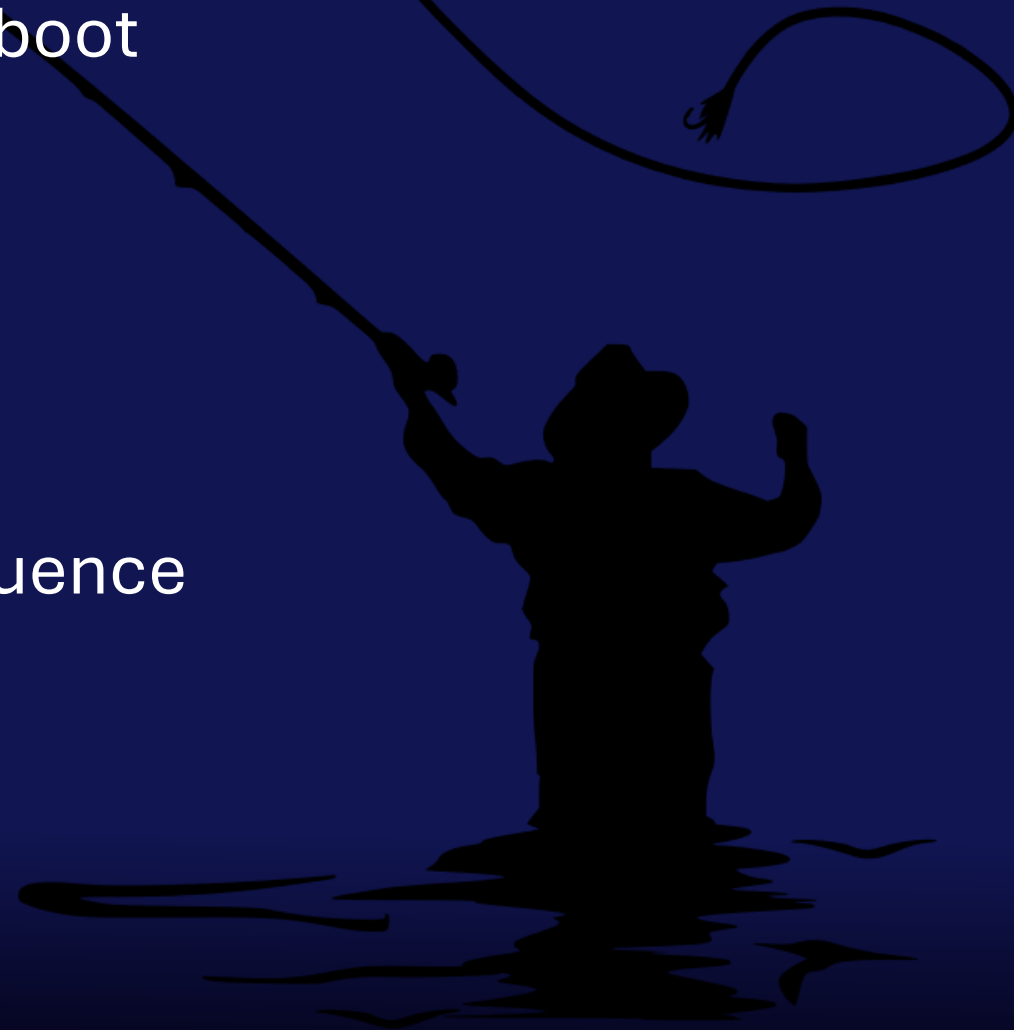


4:15 PM
5/1/2025



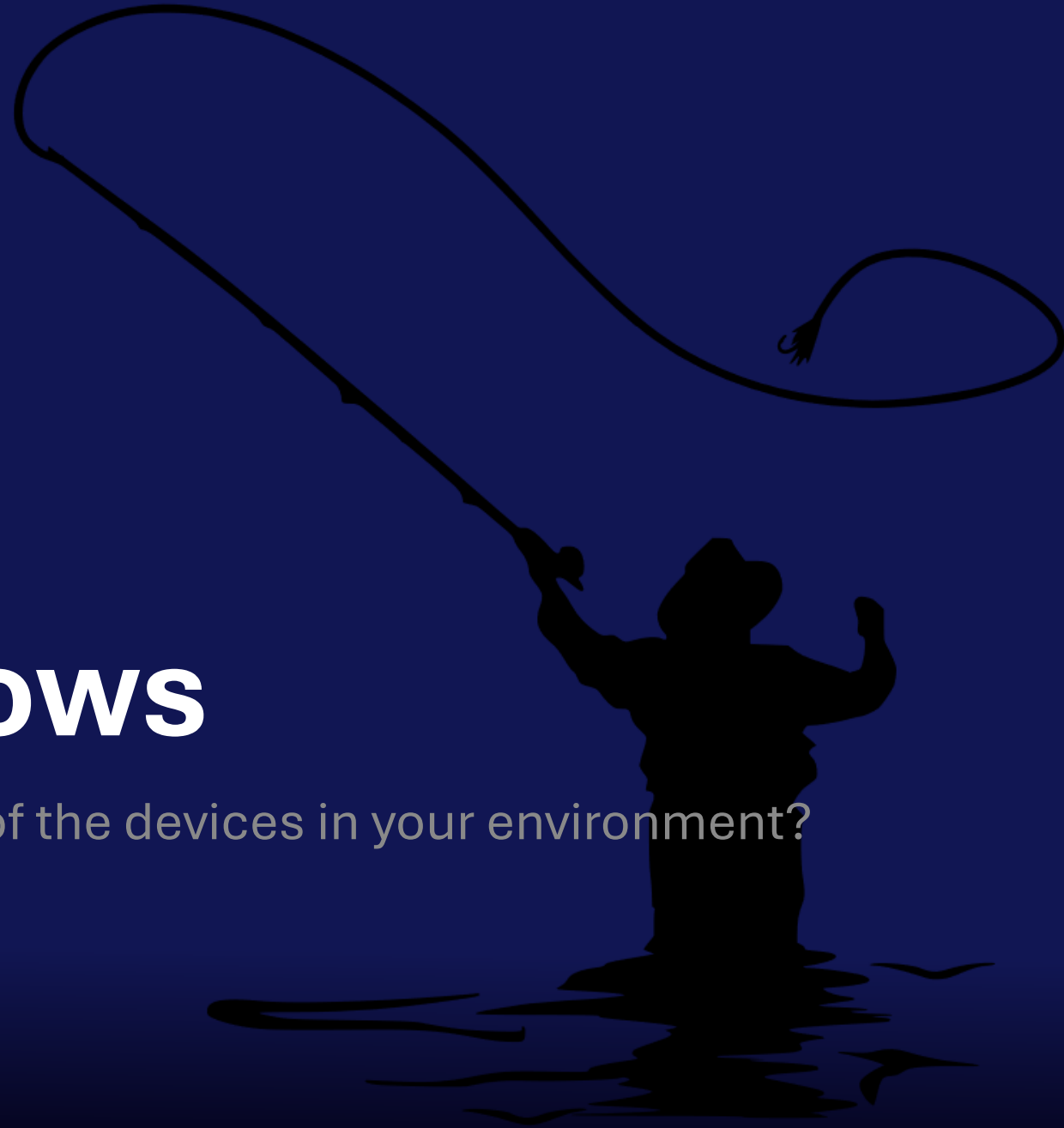
Repair Windows: Boot to (i)PXE, run fix

- ◆ OS and or hard drive may not be able to boot
- ◆ WinRE not working (or non-existent)
- ◆ Otherwise, device is still functional
- ◆ Fix is known and available
- ◆ Needs to be run from WinPE/WinRE
- ◆ (i)PXE boot device to run fix via Task Sequence



Reinstall Windows

Can you quickly and easily rebuild 10% of the devices in your environment?
25%, 50%?



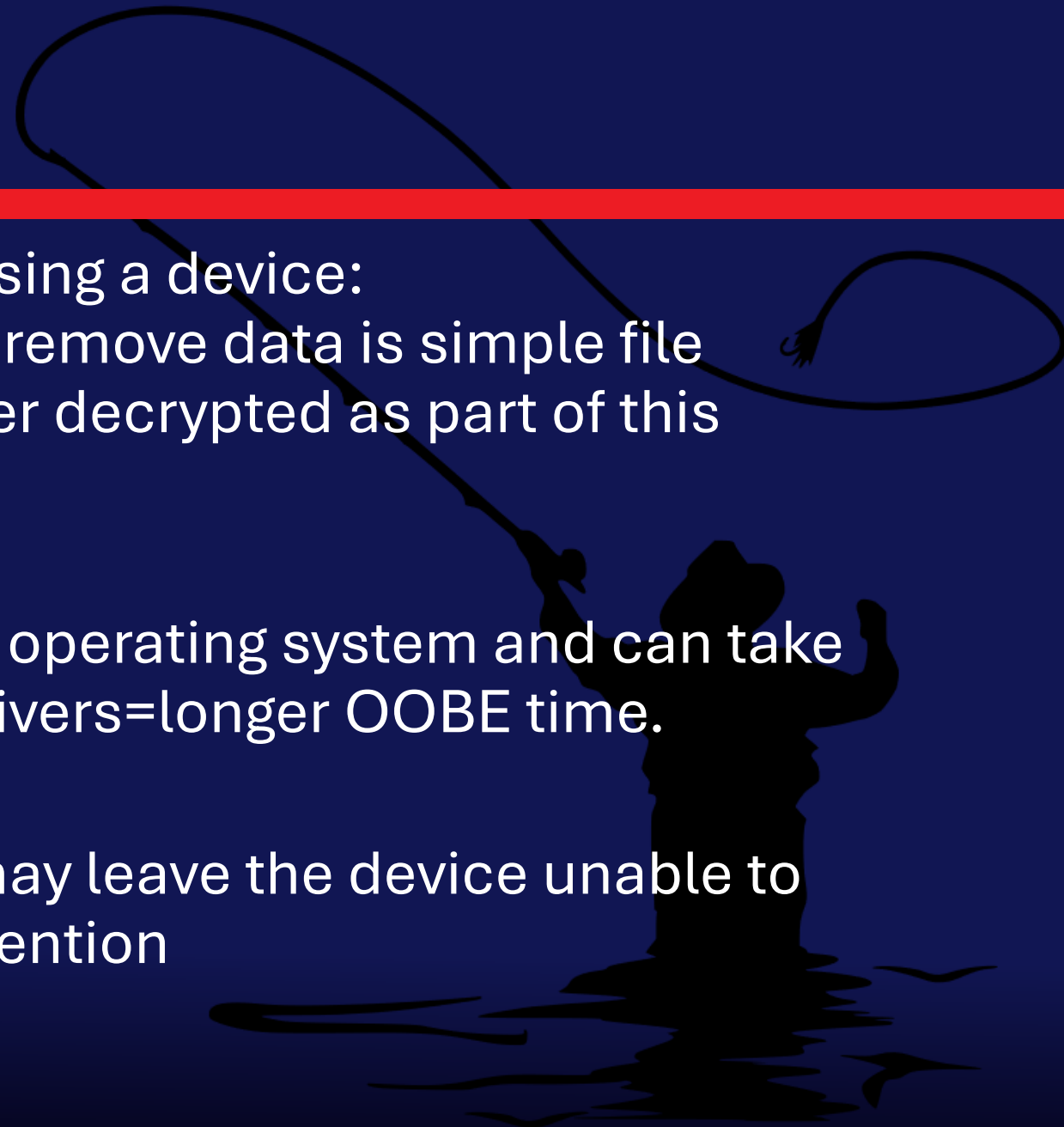
Options to reinstall Windows

- ◆ CM OSD – nah, got rid of it. MSFT said it wasn't needed anymore.
- ◆ MDT – dead and doesn't officially support Win11.
- ◆ Autopilot – nope! Autopilot is a provisioning solution, not an imaging solution.
- ◆ Intune wipe
- ◆ Windows Media Creation Tool
- ◆ OEM recovery options
- ◆ Other options



Intune Wipe

- ◆ Not a secure method for repurposing a device:
“The method that "Wipe" uses to remove data is simple file deletion, and the drive is BitLocker decrypted as part of this process.”
- ◆ Time-consuming:
The wipe involves reinstalling the operating system and can take an hour or more. Outdated OS/drivers=longer OOB time.
- ◆ Potential reboot issues:
In some cases, the wipe action may leave the device unable to reboot and require manual intervention



DEMO

Intune wipe

And WinRE



AP-2P8NCS3 - Microsoft Intune

+

←

↺

https://intune.microsoft.com/#view/Microsoft_Intune_Devices/DeviceSettingsMenuBlade/~/_/overview/mdmDeviceId/0d8a5078-a034-4dec-9570-986eb2e4ddf0

🔍

A

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🌐

Microsoft Intune admin center

🔔

⚙️

?

💬

adminmt@2pint.onmicr...
2PINT LAB (2PINT.ONMICROSO...

⏪

Home > Devices | Windows > Windows | Windows devices >

i

AP-2P8NCS3

...

🔍 Search

×

⏪

i

Overview

∨ Manage

📄 Properties

∨ Monitor

🗂️ Resource explorer

🖨️ Hardware

📦 Discovered apps

📋 Device compliance

⚙️ Device configuration

📦 App configuration

📋 Local admin password

🔑 Recovery keys

📋 User experience

🖨️ Device diagnostics

👥 Group membership

🌱 Managed Apps

📋 Filter evaluation

📋 Enrollment

Add or remove favorites by pressing Ctrl+Shift+F

✕ Retire

↺ Wipe

🗑️ Delete

🔒 Remote lock

↺ Sync

🔑 Reset passcode

🔄 Restart

⬇️ Collect diagnostics

↺ Fresh Start

⋮

∧ Essentials

Device name
AP-2P8NCS3

Management name
adminmt_Windows_4/23/2025_3:40 AM

Ownership
Corporate

Serial number
2P8NCS3

Phone number

Device manufacturer
Dell Inc.

Primary user
[Mike Terrill \(Admin\)](#)

Enrolled by
[Mike Terrill \(Admin\)](#)

Compliance
Compliant

Operating system
Windows

Device model
Latitude 3420

Last check-in time
4/23/2025, 10:19:19 AM

Remote assistance
Not configured

Device actions status

Action	Status	Date/Time	Error
No data			

Activate Windows

Go to Settings to activate Windows.

84°F
Sunny

🔍 Search

📁

📁

⬆️

📶

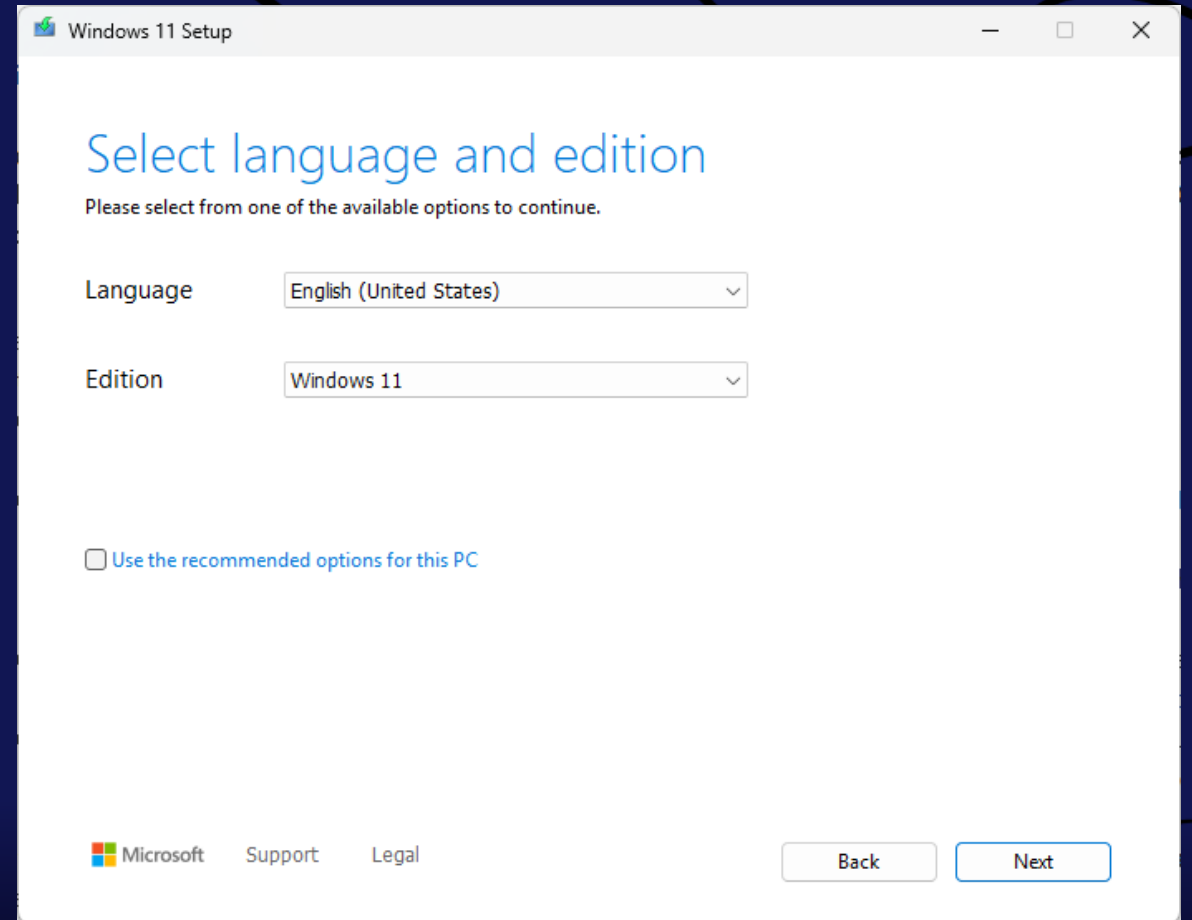
🔊

🔌

3:56 PM
4/23/2025

Windows Media Creation Tool

- ◆ For advanced end users only
- ◆ Cannot select version
- ◆ Ability to select language
- ◆ Doesn't require a recovery partition
- ◆ Out-of-box drivers only
- ◆ Requires a USB key (8GB+)
- ◆ Additional setup required
- ◆ Not automated
- ◆ Not scalable



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MICROSOFT SOFTWARE LICENSE TERMS

MICROSOFT MEDIA CREATION TOOL

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Support

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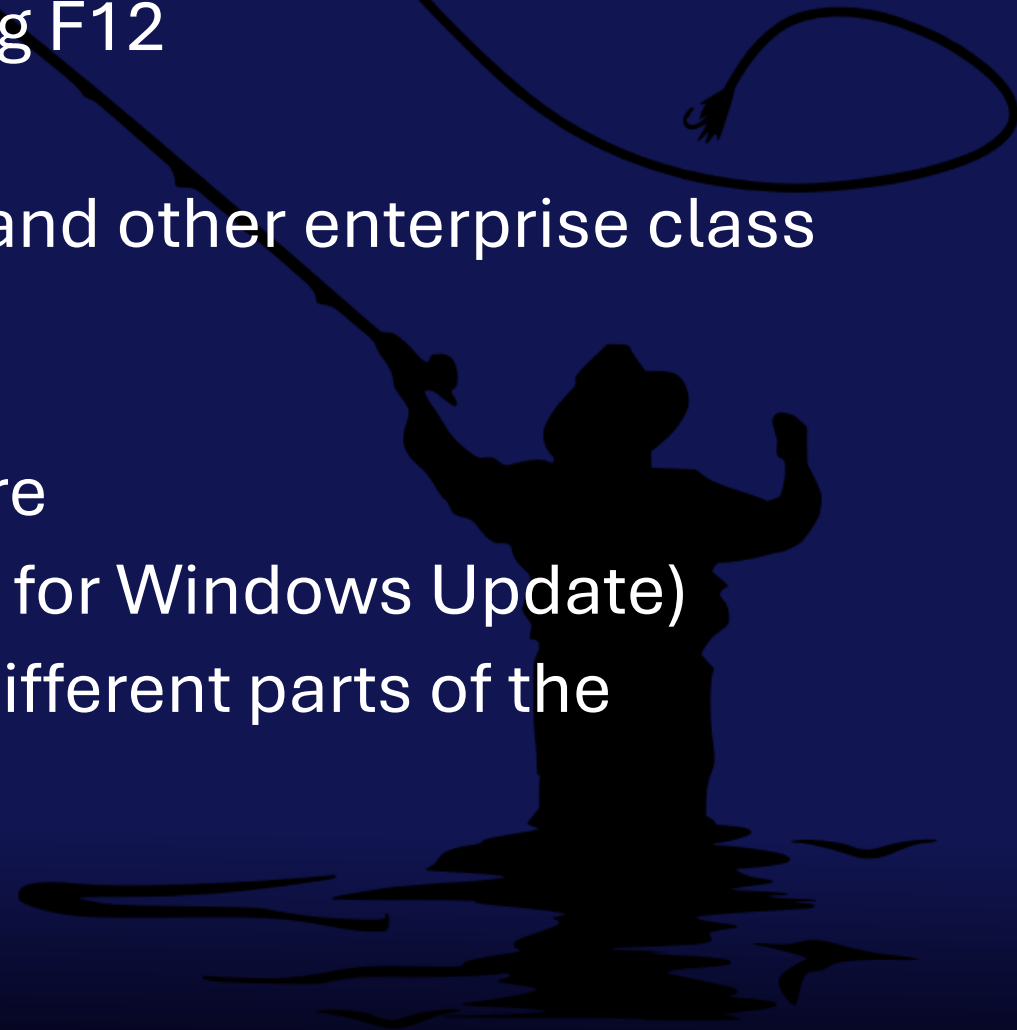
MMS

Dell Windows Recovery Options



Dell SupportAssist OS Recovery

- ◆ Boot to SupportAssist OS Recovery using F12
- ◆ Requires intact Dell recovery partition
- ◆ Works on Precision, Latitude, OptiPlex, and other enterprise class models
- ◆ Works on Wi-Fi*
- ◆ Includes Dell factory drivers and software
- ◆ Outdated OS/Drivers (longer OOBE time for Windows Update)
- ◆ Not fully automated - Input required at different parts of the process
- ◆ Not scalable





Mike Terrill
adminmt@2pint.onmicrosoft.com



Home



System



Bluetooth & devices



Network & internet



Personalization



Apps



Accounts



Time & language



Gaming



Accessibility



Privacy & security



Windows Update

Windows Update



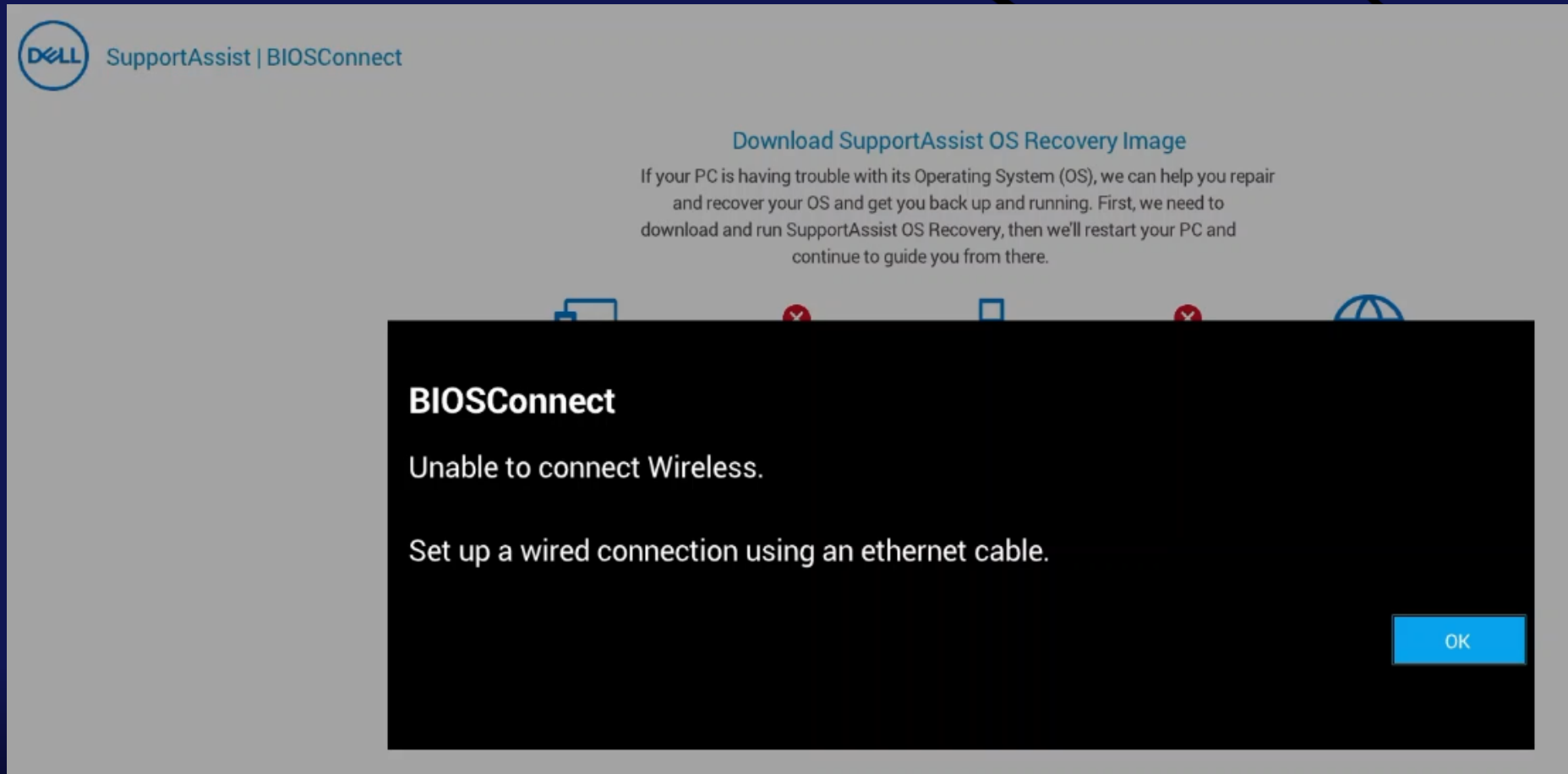
Updates available

Last checked: Today, 5:53 PM

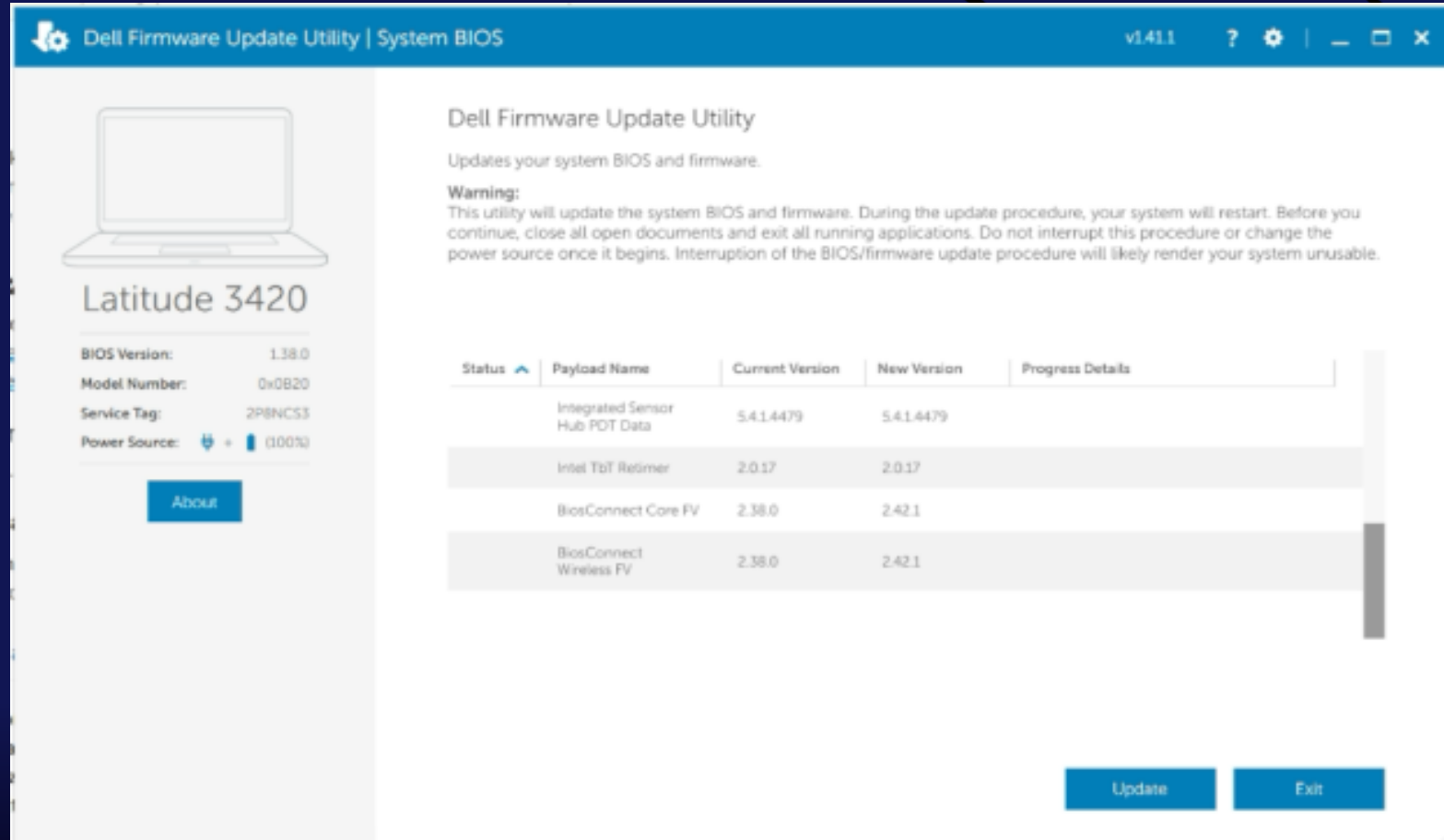
[Download & install all](#)

Security Intelligence Update for Microsoft Defender Antivirus - KB2267602 (Version 1.431.414.0) - Current Channel (Broad)	Downloading - 0%
2025-06 Cumulative Update for Windows 11 Version 24H2 for x64-based Systems (KB5063060)	Downloading - 0%
Intel - net - 23.110.0.5	Downloading - 0%
Windows Malicious Software Removal Tool x64 - v5.134 (KB890830)	Downloading - 0%
Update for Windows Security platform - KB5007651 (Version 10.0.27840.1000)	Downloading - 0%
Dell, Inc. - Firmware - 11.20.1.9	Downloading - 0%
2025-04 Cumulative Update for .NET Framework 3.5 and 4.8.1 for Windows 11, version 24H2 for x64 (KB5054979)	Downloading - 0%
Intel Corporation - Display - 32.0.101.6078	Downloading - 0%
Intel Corporation - SoftwareComponent - 20.0.0.1038	Downloading - 0%
Intel Corporation - SCSIAdapter - 20.0.0.1038	Downloading - 0%
Intel Corporation - Extension - 20.0.0.1038	Downloading - 0%
Intel - Net - 12.19.2.56	Downloading - 0%
Dell, Inc. - Firmware - 0.1.2.0	Downloading - 0%

Dell SupportAssist – Missing Wi-Fi



Dell SupportAssist – BIOS Update



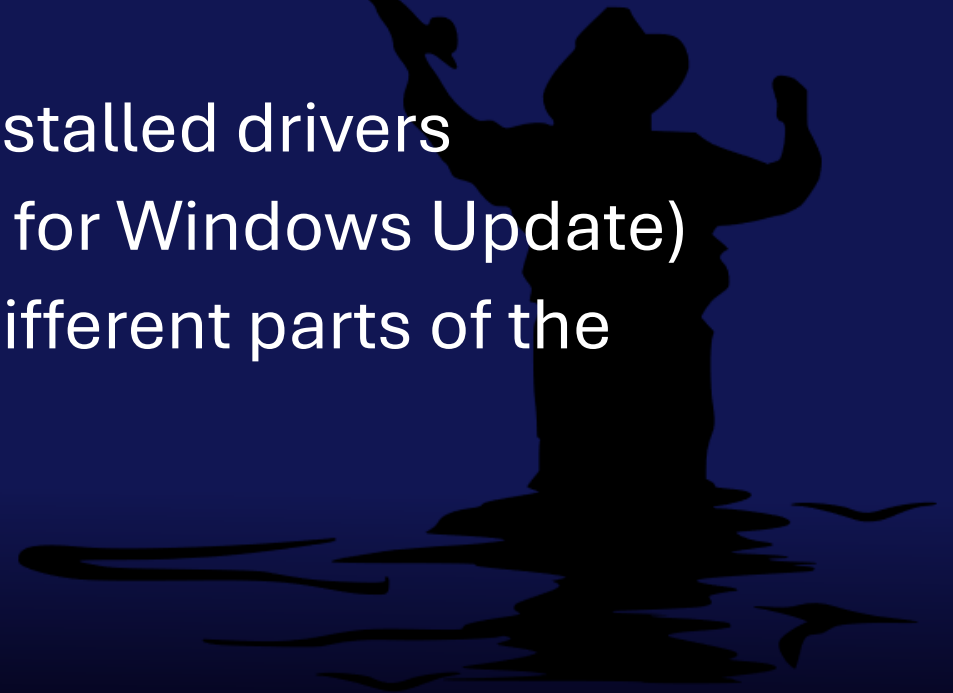
The screenshot shows the Dell Firmware Update Utility window. The title bar reads "Dell Firmware Update Utility | System BIOS" with version "v1.41.1" and standard window controls. On the left, a laptop icon is above the model name "Latitude 3420". Below this, system information is listed: BIOS Version: 1.38.0, Model Number: 0x0820, Service Tag: 2P8NC53, and Power Source: 100%. An "About" button is at the bottom of this section. The main area is titled "Dell Firmware Update Utility" and contains a warning: "This utility will update the system BIOS and firmware. During the update procedure, your system will restart. Before you continue, close all open documents and exit all running applications. Do not interrupt this procedure or change the power source once it begins. Interruption of the BIOS/firmware update procedure will likely render your system unusable." Below the warning is a table showing update details for various components.

Status	Payload Name	Current Version	New Version	Progress Details
	Integrated Sensor Hub PDT Data	5.4.1.4479	5.4.1.4479	
	Intel TbT Retimer	2.0.17	2.0.17	
	BiosConnect Core FV	2.38.0	2.42.1	
	BiosConnect Wireless FV	2.38.0	2.42.1	

At the bottom right, there are "Update" and "Exit" buttons.

Dell OS Recovery Tool (USB)

- ◆ Download the Dell OS Recovery Tool (follows same process as SupportAssist)
- ◆ Requires Service tag of target Dell PC and USB key (16GB+)
- ◆ Works on Precision, Latitude, OptiPlex, and other enterprise class models
- ◆ Restores exact factory image with pre-installed drivers
- ◆ Outdated OS/Drivers (longer OOB time for Windows Update)
- ◆ Not fully automated - Input required at different parts of the process
- ◆ Not scalable

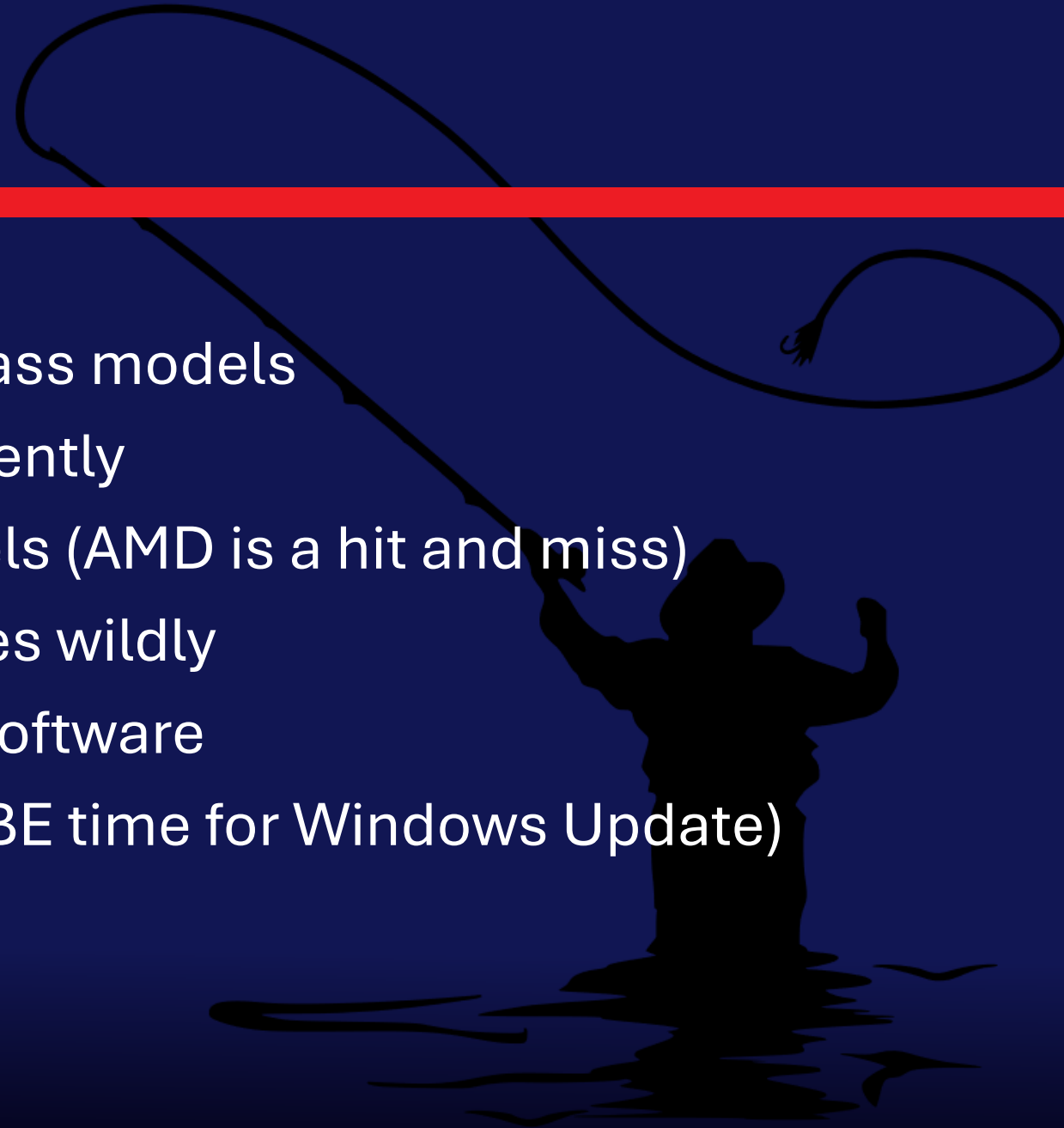


HP Windows Recovery Options



HP Sure Recover

- ◆ Boot to Sure Recover using F11
- ◆ Works on higher end business class models
- ◆ Make sure BIOS is updated frequently
- ◆ Wi-Fi is available on recent models (AMD is a hit and miss)
- ◆ Speeds, times, and success varies wildly
- ◆ Includes HP factory drivers and software
- ◆ Outdated OS/Drivers (longer OOBE time for Windows Update)
- ◆ Not scalable



Windows 11

Microsoft Windows
Version 24H2 (OS Build 26100.1742)
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temp
org name

OK

Complete updates to help keep your device secure

Your device is missing important security updates. Make sure to keep your device complete.

Security Intelligence Update for Microsoft Defender Antivirus - KB2267602 (Version 1.431.497.0)

2025-07 Cumulative Update for .NET Framework 3.5 and 4.8.1 for Windows 11, version 24H2 for

2025-07 Cumulative Update for Windows 11 Version 24H2 for x64-based Systems (KB5062553)

HP Development Company, L.P. - SoftwareComponent - 8.10.29.1

HP Inc. - SoftwareComponent - 1.79.4265.0

HP Inc. - Extension - 4.8.85.0

HP Development Company, L.P. - Keyboard - 11.1.9.1

HP - Firmware - 1.9.0.0

HP Development Company, L.P. - Extension - 8.10.29.1

Windows Malicious Software Removal Tool x64 - v5.134 (KB890830)

HP Inc. - SoftwareDevice - 1.1.6.47

Update for Windows Security platform - KB5007651 (Version 10.0.27840.1000)

Intel - net - 23.120.0.3

HP Inc. - SoftwareComponent - 4.2.2439.0

Intel Corporation - Bluetooth - 23.80.0.3

Intel - Extension - 9.0.11701.44281

Organize

Name	Publisher	Installed On	Size	Version
HP Documentation	HP Inc.	7/9/2025	18.1 MB	1.0.0.1
HP Notifications	HP	7/9/2025		1.1.31.56
Microsoft 365 Apps for enterprise - en-gb	Microsoft Corporation	12/5/2024	420 MB	16.0.17328.20206
Microsoft 365 Apps for enterprise - en-us	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 Apps for enterprise - fr-ca	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 Apps for enterprise - fr-fr	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 Apps for enterprise - ja-jp	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 Apps for enterprise - th-th	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 Apps para Grandes Empresas - pt-br	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 Apps 企業版 - zh-tw	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft 365 企業應用版 - zh-cn	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft Edge	Microsoft Corporation	7/9/2025		138.0.3351.77
Microsoft OneDrive	Microsoft Corporation	7/9/2025	420 MB	25.110.0608.0002
Microsoft OneNote - en-gb	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - en-us	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - es-es	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - es-mx	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - fr-ca	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - fr-fr	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - ja-jp	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - ko-kr	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - pt-br	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - th-th	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - zh-cn	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft OneNote - zh-tw	Microsoft Corporation	12/5/2024		16.0.17328.20206
Microsoft Visual C++ 2005 Redistributable	Microsoft Corporation	7/9/2025	4.99 MB	8.0.56336
Microsoft Visual C++ 2005 Redistributable (x64)	Microsoft Corporation	7/9/2025	7.00 MB	8.0.56336
Microsoft Visual C++ 2008 Redistributable - x64 9...	Microsoft Corporation	7/9/2025	13.2 MB	9.0.30729
Microsoft Visual C++ 2008 Redistributable - x86 9...	Microsoft Corporation	7/9/2025	10.1 MB	9.0.30729.4148
Microsoft Visual C++ 2010 x64 Redistributable - 1...	Microsoft Corporation	7/9/2025	13.8 MB	10.0.40219
Microsoft Visual C++ 2010 x86 Redistributable - 1...	Microsoft Corporation	7/9/2025	11.1 MB	10.0.40219
Microsoft Visual C++ 2012 Redistributable (x64) - ...	Microsoft Corporation	7/9/2025	20.5 MB	11.0.51106.1
Microsoft Visual C++ 2012 Redistributable (x86) - ...	Microsoft Corporation	7/9/2025	17.4 MB	11.0.51106.1
Microsoft Visual C++ 2013 Redistributable (x64) - ...	Microsoft Corporation	7/9/2025	20.5 MB	12.0.30501.0
Microsoft Visual C++ 2013 Redistributable (x86) - ...	Microsoft Corporation	7/9/2025	17.1 MB	12.0.30501.0
Microsoft Visual C++ 2015 Redistributable (x64) - ...	Microsoft Corporation	7/9/2025	22.4 MB	14.0.23026.0
Microsoft Visual C++ 2015 Redistributable (x86) - ...	Microsoft Corporation	7/9/2025	18.6 MB	14.0.23026.0
Remote Desktop Connection	Microsoft Corporation	12/5/2024		
엔터프라이즈용 Microsoft 365 앱 - ko-kr	Microsoft Corporation	12/5/2024		16.0.17328.20206

Currently installed programs Total size: 616 MB

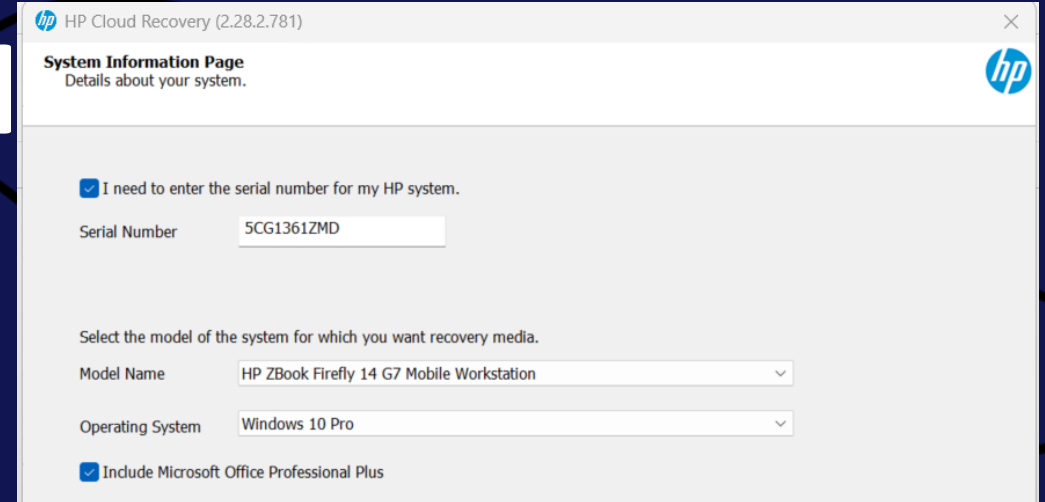
HP Cloud Recovery Tool

- ◆ Download the HP Cloud Recovery Tool
- ◆ Requires serial number (or run on the device) and USB key (32GB)
- ◆ Includes HP factory drivers and software
- ◆ Takes forever – type of flash drive matters
- ◆ Outdated OS/Drivers (longer OOB time for Windows Update)
- ◆ Not scalable



HP Cloud Recovery Tool

- Requires
 - 32GB Flash Drive
 - Run on machine you want image for, or Serial Number
- Pros:
 - Includes HP factory drivers and software
- Cons:
 - Box of Chocolates....
 - Outdated OS/
 - Not scalable
 - Takes FOREVER | Flash Drives Matter



HP Cloud Recovery (2.28.2.781)

System Information Page
Details about your system.

☒ I need to enter the serial number for my HP system.

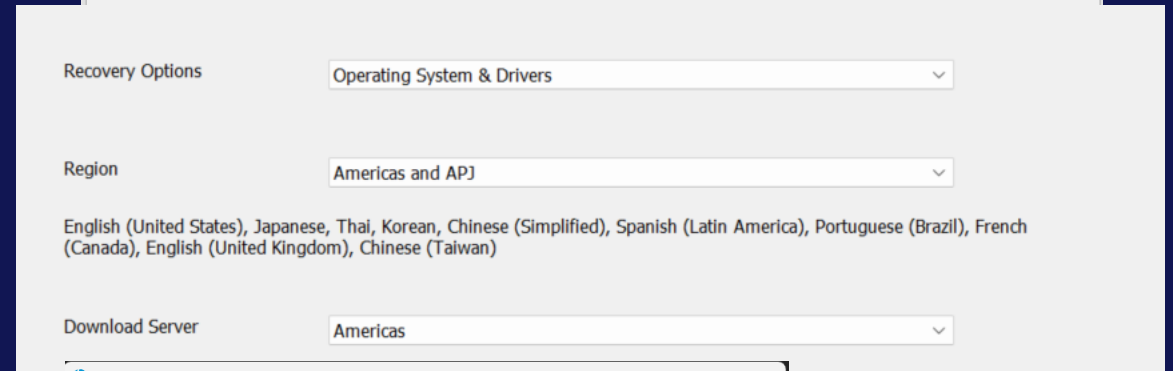
Serial Number

Select the model of the system for which you want recovery media.

Model Name

Operating System

☒ Include Microsoft Office Professional Plus

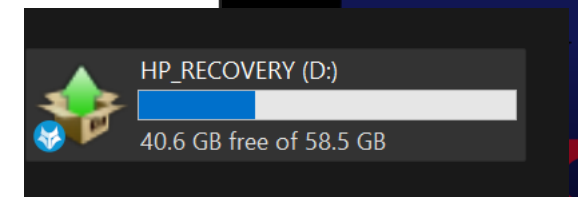
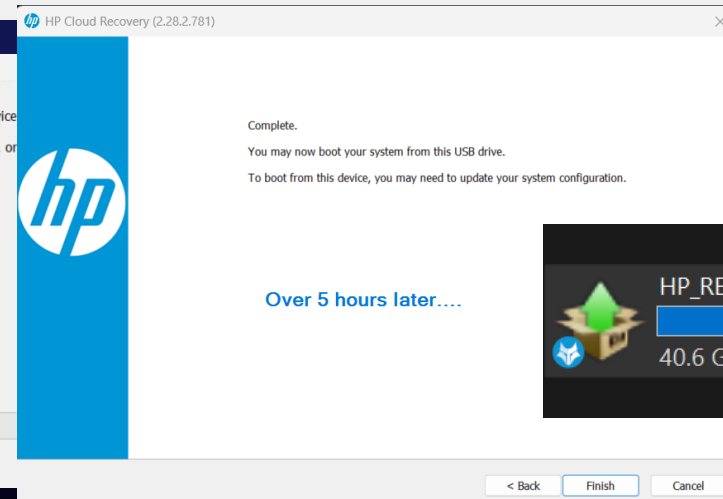
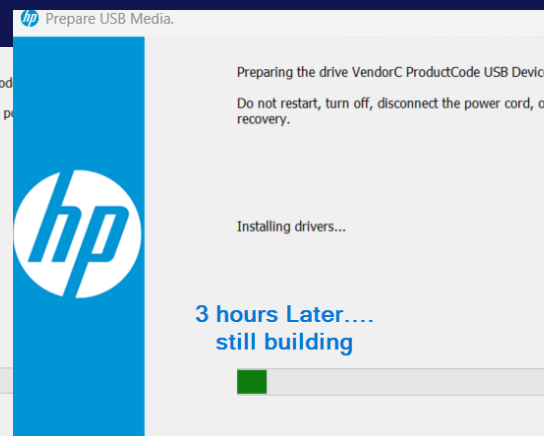
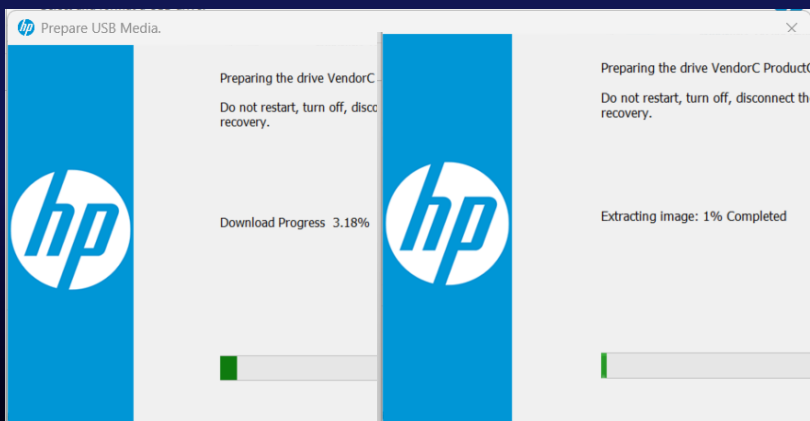


Recovery Options

Region

English (United States), Japanese, Thai, Korean, Chinese (Simplified), Spanish (Latin America), Portuguese (Brazil), French (Canada), English (United Kingdom), Chinese (Taiwan)

Download Server



Lenovo Windows Recovery Options



Lenovo USB Recovery Creator

Sign in

We've built a new experience allowing you to keep track of your Lenovo products and orders, all in one place. [Create a Lenovo ID.](#)

 E-mail address *

 Password *

 Verification Code *

05266

☐ Remember me

[Forgot Password](#)

SIGN IN

Or Sign In With



Facebook



Google



Live ID

Lenovo Recovery Media



Determine Eligibility



Add Order



Customer Info



Review & Confirm

[< Back](#)

Order Details

Part Number: 5R31N66991



Product Name	NB TP X1 Carbon G11 I7 32G 1T 11P
Serial Number	PF4L9NC8
Machine Type	21HM
Country / Region	United States Of America

Operating System	Win11PROx64
OS Language	English

[My preferred OS language is not listed.](#)

Select Media Type

☐ USB Media

☒ Digital Download Recommended

☒ I accept the [Lenovo License Agreement](#) & [Lenovo Software Privacy Statement](#)

Size: 32G Price: \$149.99

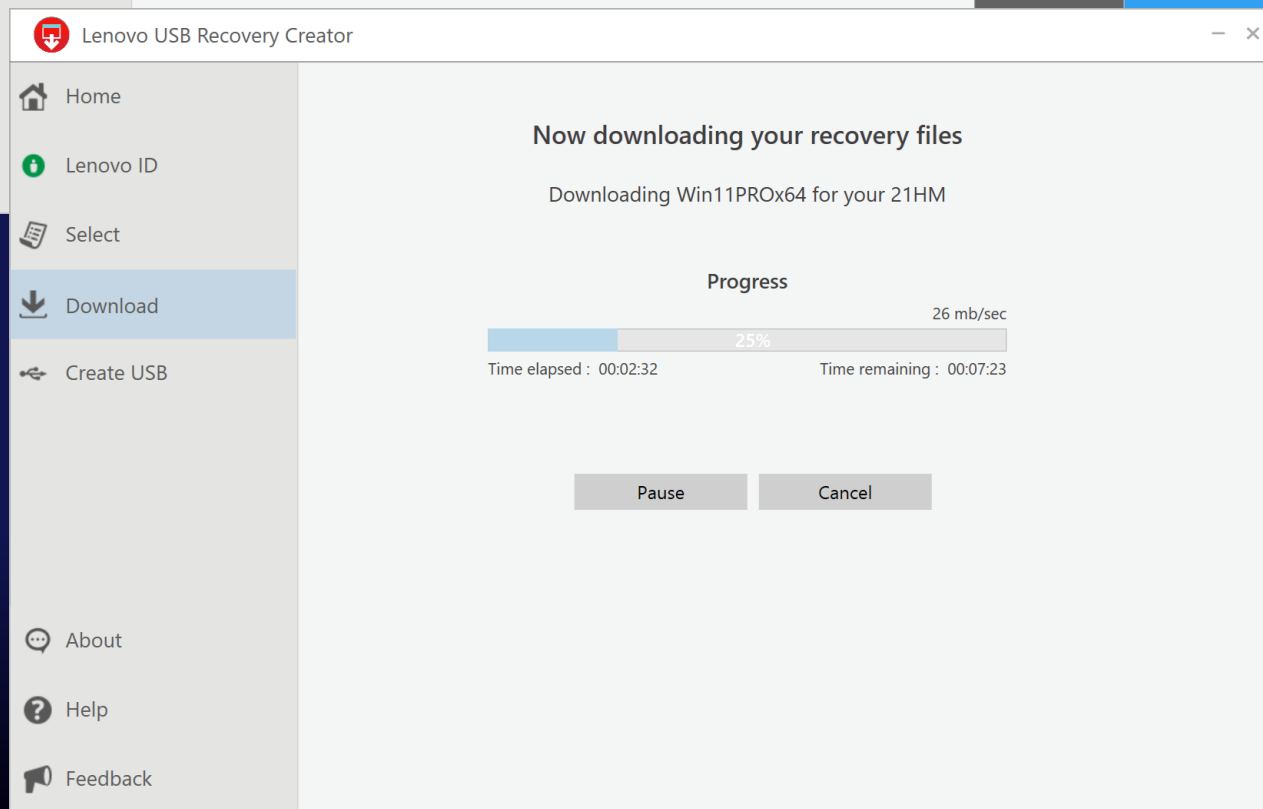
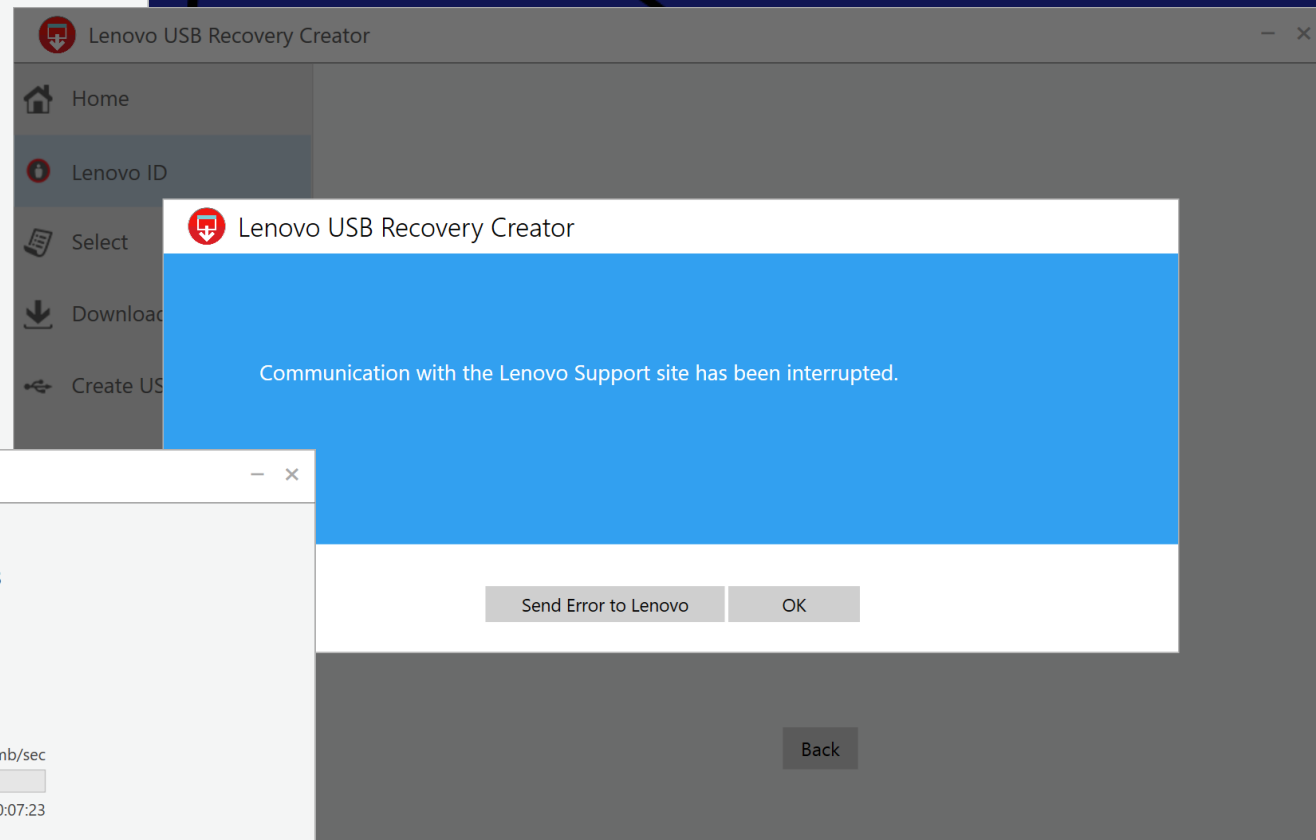
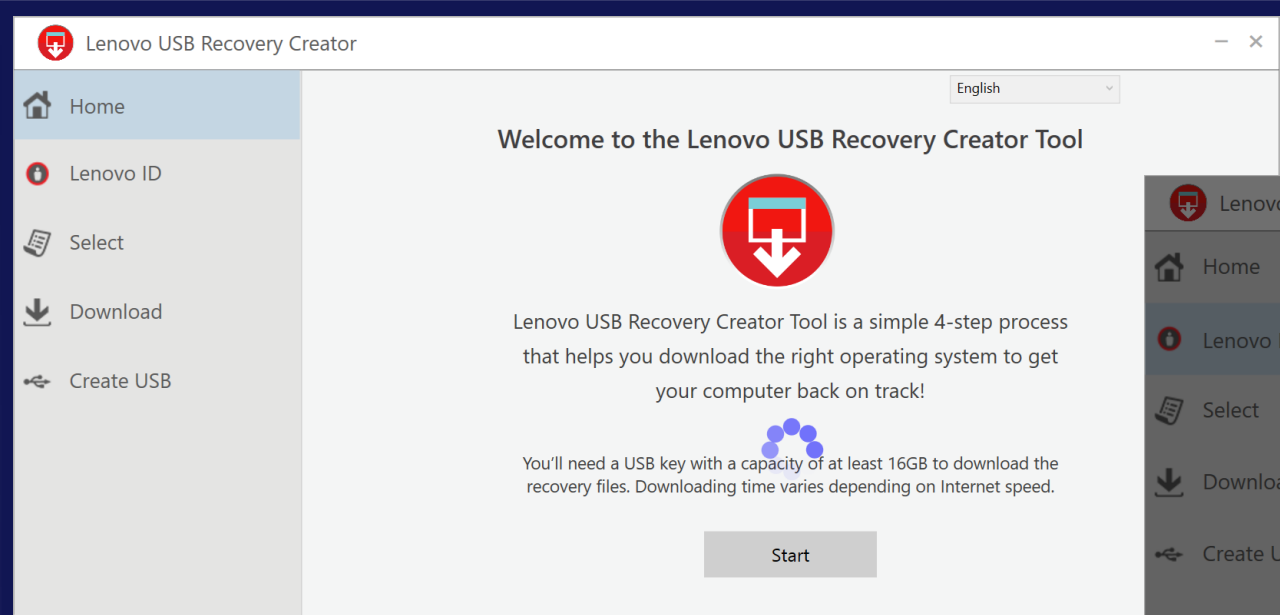
Cancel

Next

You will also receive an email confirmation, if assistance is needed, please send email to recovery@lenovo.com.

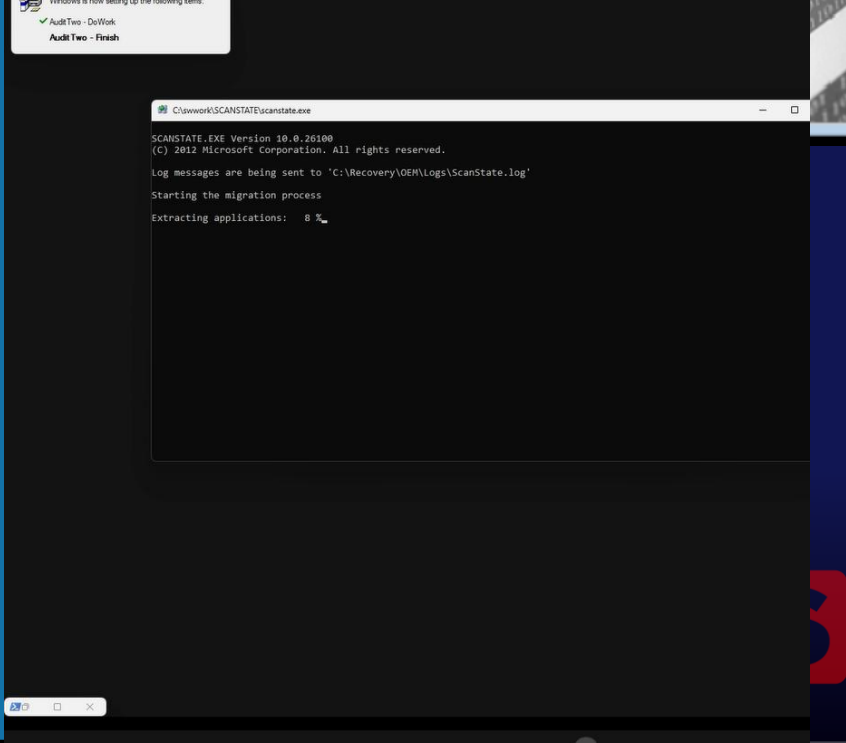
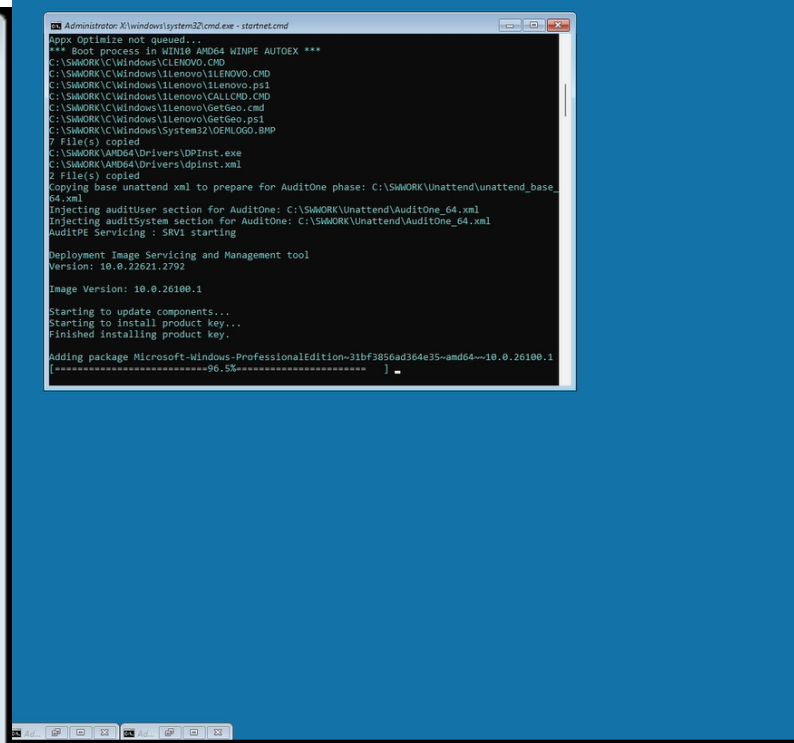
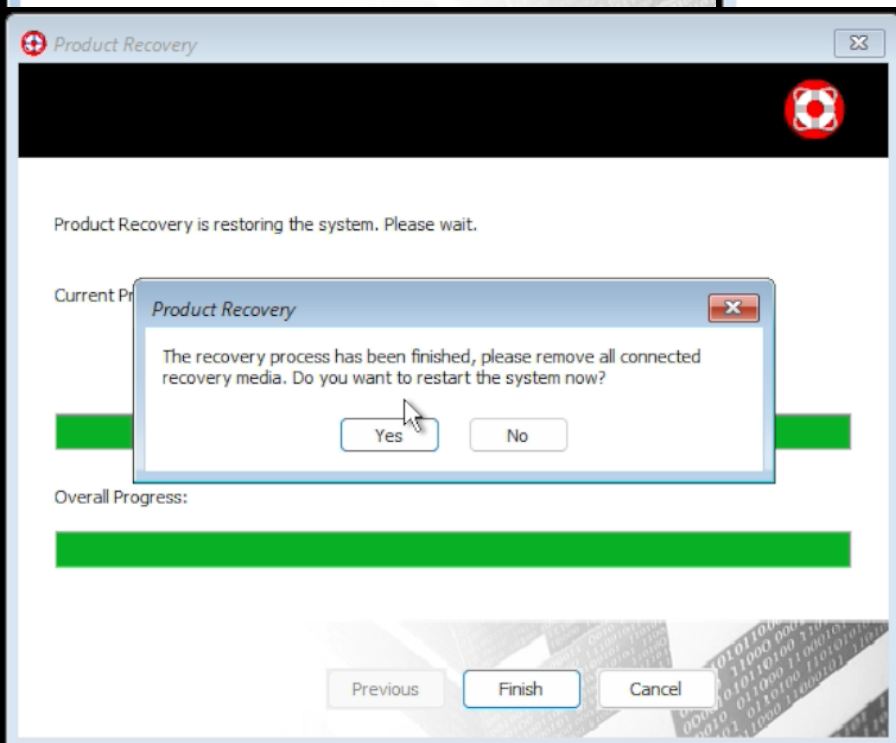
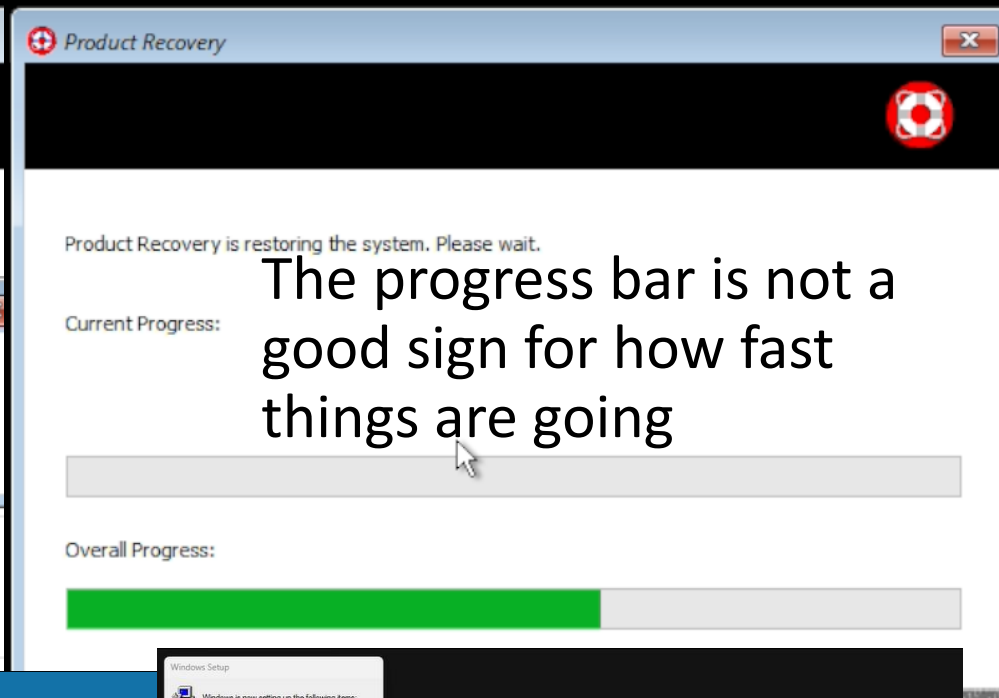
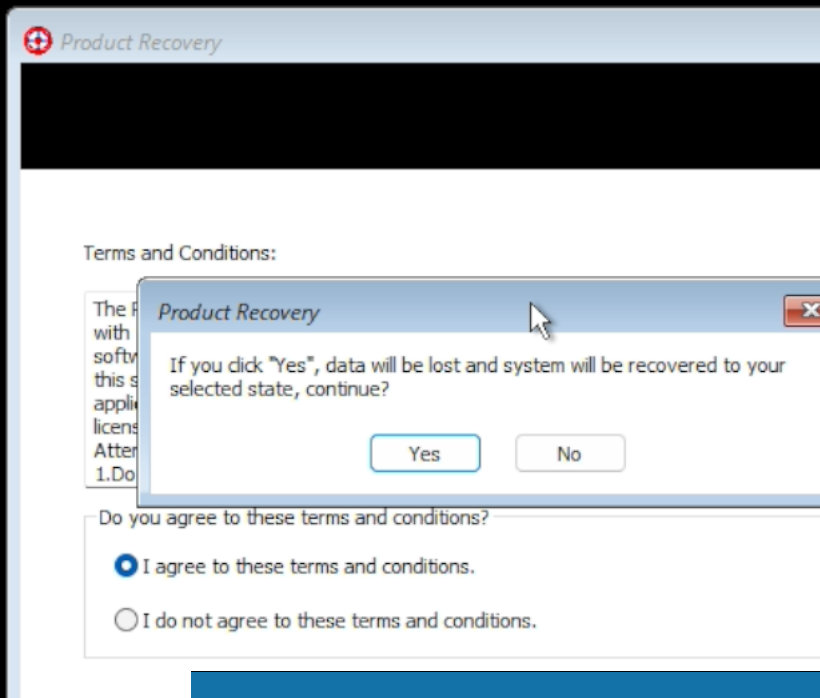
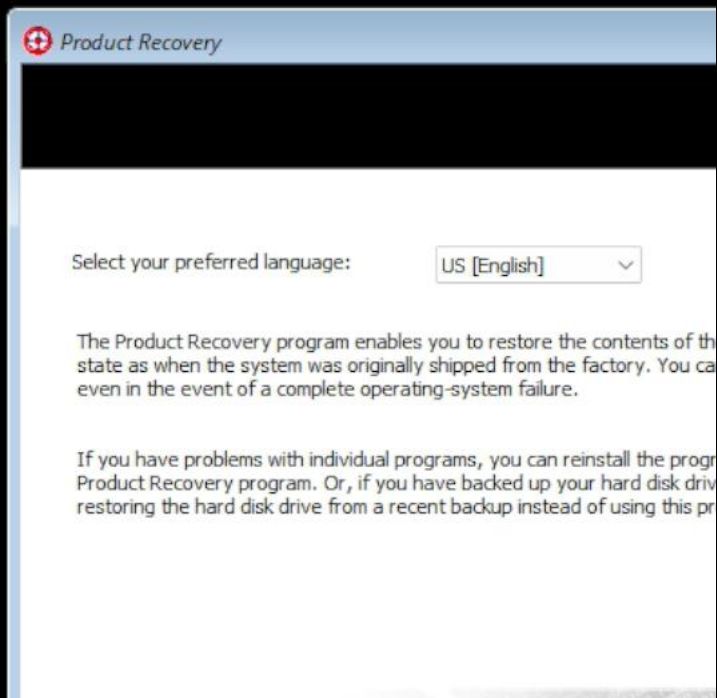
Note: A digital download order is active for 72 hours (3 days). If a digital download order has not started within 72 hours, the digital download order expires, and a new digital download order must be submitted. When a digital download order has started, it will remain active until the digital download has completed.

MMS



Apparently, if you're blocking traffic in/out from China, this tool doesn't work so well.





About Windows

Windows 11

Microsoft Windows

Version 24H2 (OS Build 26100.1)

© Microsoft Corporation. All rights reserved.

The Windows 11 Pro operating system and its user interface are protected by trademark and other pending or existing intellectual property rights in the United States and other countries/regions.

This product is licensed under the [Microsoft Software License Terms](#) to:

Lenovo

org name

OK

Type here to search

Pinned

Edge

Photos

Calculator

Get S
Welc

Brand Guidelines
2h ago

Word

Excel

Microsoft Store

Settings

More >

Edit name

Create folders in Start

Keep what's important close by grouping your pinned apps into folders.

• • • • •

Your update is in progress—this could take a while. Please keep your PC on and plugged in.

Step 1 of 3: Downloading - 98%

Organize				
Name	Publisher	Installed On	Size	Version
Lenovo Vantage Service	Lenovo Group Ltd.	7/10/2025	30.3 MB	4.0.52.0
Microsoft 365 - en-us	Microsoft Corporation	7/10/2025		16.0.17328.20206
Microsoft Edge	Microsoft Corporation	7/10/2025		138.0.3351.77
Microsoft OneNote - en-us	Microsoft Corporation	7/10/2025		16.0.17328.20206
Microsoft Visual C++ 2015-2022 Redistributable (x...	Microsoft Corporation	7/10/2025	20.1 MB	14.30.30704.0
Realtek USB Audio	Realtek Semiconductor Corp.	7/10/2025		6.3.9600.250
Remote Desktop Connection	Microsoft Corporation	5/23/2024		

Windows Update



Updates available to download

Last checked: Today, 4:55 PM

Download & install all

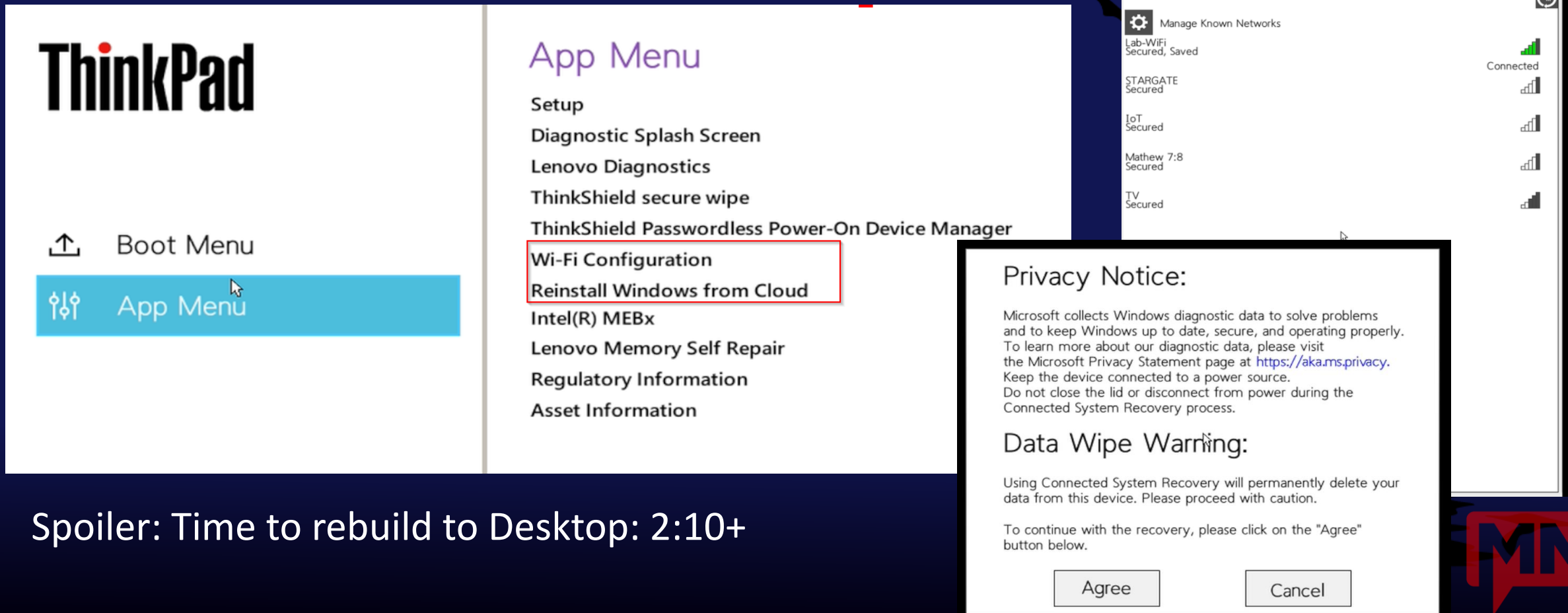
Security Intelligence Update for Microsoft Defender Antivirus - KB2267602 (Version 1.431.523.0) - Current Channel (Broad)	Download & install
Lenovo - System - 2.2.30.0	Download & install
Lenovo - System - 10.14.1.4	Download & install
Intel Corporation - Display - 32.0.101.6733	Download & install
2025-07 Cumulative Update for .NET Framework 3.5 and 4.8.1 for Windows 11, version 24H2 for x64 (KB5056579)	Download & install
Intel - SoftwareComponent - 2441.7.0.0	Download & install
Intel - SoftwareComponent - 2511.7.5.0	Download & install
Intel - Ports - 2441.7.0.0	Download & install
Dolby - Extension - 9.730.1223.7	Download & install
Dolby - Extension - 9.626.711.4	Download & install
Realtek - SoftwareComponent - 11.0.6000.353	Download & install
Realtek Semiconductor Corp. - MEDIA - 6.0.9764.1	Download & install
Realtek - SoftwareComponent - 1.0.810.0	Download & install
Dolby - Extension - 9.703.719.14	Download & install

Over 2 Hours to get to Desktop, and still
WU had a lot of updates to do.
24H2 – 26100.1



Lenovo – Reinstall Windows from Cloud

- ◆ BIOS: Network -> Reinstall Windows from Cloud -> Enable
- ◆ F12 Boot -> Setup WiFi | Trigger Process



The image shows a composite of three screenshots from a Lenovo ThinkPad recovery environment. On the left is the 'ThinkPad' logo and a 'Boot Menu' with 'App Menu' selected. The middle screenshot shows the 'App Menu' with 'Reinstall Windows from Cloud' highlighted in a red box. On the right is the 'WiFi Manager' window showing a list of networks, with 'Lab-WiFi' selected. In the foreground, a 'Privacy Notice' and 'Data Wipe Warning' dialog box is displayed, with 'Agree' and 'Cancel' buttons at the bottom.

ThinkPad

Boot Menu

App Menu

App Menu

- Setup
- Diagnostic Splash Screen
- Lenovo Diagnostics
- ThinkShield secure wipe
- ThinkShield Passwordless Power-On Device Manager
- Wi-Fi Configuration**
- Reinstall Windows from Cloud**
- Intel(R) MEBx
- Lenovo Memory Self Repair
- Regulatory Information
- Asset Information

WiFi Manager - 88:F4:DA:D7:68:0B

- Manage Known Networks
- Lab-WiFi Secured, Saved
- STARGATE Secured
- IoT Secured
- Mathew 7:8 Secured
- TV Secured

Privacy Notice:

Microsoft collects Windows diagnostic data to solve problems and to keep Windows up to date, secure, and operating properly. To learn more about our diagnostic data, please visit the Microsoft Privacy Statement page at <https://aka.ms/privacy>. Keep the device connected to a power source. Do not close the lid or disconnect from power during the Connected System Recovery process.

Data Wipe Warning:

Using Connected System Recovery will permanently delete your data from this device. Please proceed with caution.

To continue with the recovery, please click on the "Agree" button below.

Agree Cancel

Spoiler: Time to rebuild to Desktop: 2:10+



Le

.

Installing 33%

Please keep your computer on and plugged in.

The preparation for
may take several minutes.



.

Preparing

Your computer may restart a few times.

Computer's TPM (Trusted Platform Module).

1. You will lose all created keys and

**REQUIRES USER TO
CONFIRM**

Press [F9] to confirm, ESC to reject.

MMS



- Home
- System
- Bluetooth
- Network
- Personalization
- Applications
- Accounts
- Time & Language
- Windows Defender

Device Manager

File Action View Help

DESKTOP-GEFCUCO

- > Audio inputs and outputs
- > Audio Processing Objects (APOs)
- > Batteries
- > Biometric devices
- > Bluetooth
- > Cameras
- > Computer
- > Disk drives
- > Display adapters
- > Firmware
- > Human Interface Devices
- > Keyboards
- > Mice and other pointing devices
- > Monitors
- > Network adapters
- > Neural processors
- > Portable Devices
- > Ports (COM & LPT)
- > Print queues
- > Processors
- > Security devices
- > Sensors
- > Software components
- > Software devices
- > Sound, video and game controllers
- > Storage controllers
- > System devices
- > Universal Serial Bus controllers
- > Universal Serial Bus devices
- > USB Connector Managers

Windows Update



Updates available

Last checked: Today, 11:44 PM

Download & install all

2025-07 Cumulative Update Preview for Windows 11 Version 24H2 for x64-based Systems (KB5062660)	Downloading - 2%
Security Intelligence Update for Microsoft Defender Antivirus - KB2267602 (Version 1.431.536.0) - Current Channel (Broad)	Download & install
Windows Malicious Software Removal Tool x64 - v5.134 (KB890830)	Download & install
Update for Windows Security platform - KB5007651 (Version 10.0.27840.1000)	Download & install
Intel - SoftwareComponent - 1.0.0.49	Download & install
Intel - SoftwareComponent - 9.0.11702.46642	Download & install
Microsoft Corporation - SoftwareComponent - 2.0.15.0	Download & install

Programs and Features

Control Panel > Programs > Programs and Features

Control Panel Home

Uninstall or change a program

To uninstall a program, select it from the list and then click Uninstall, Change, or Repair.

Name	Publisher	Installed On	Size	Version
Microsoft Edge	Microsoft Corporation	7/10/2025		138.0.3351.83
Microsoft Visual C++ 2015-2022 Redistributable (x64) ...	Microsoft Corporation	7/11/2025	20.1 MB	14.30.30704.0
Realtek USB Audio	Realtek Semiconductor Corp.	7/11/2025		6.3.9600.250
Remote Desktop Connection	Microsoft Corporation	7/11/2025		

Currently installed programs Total size: 20.1 MB
4 programs installed

Panasonic Windows Recovery Options



Panasonic Recovery Partition

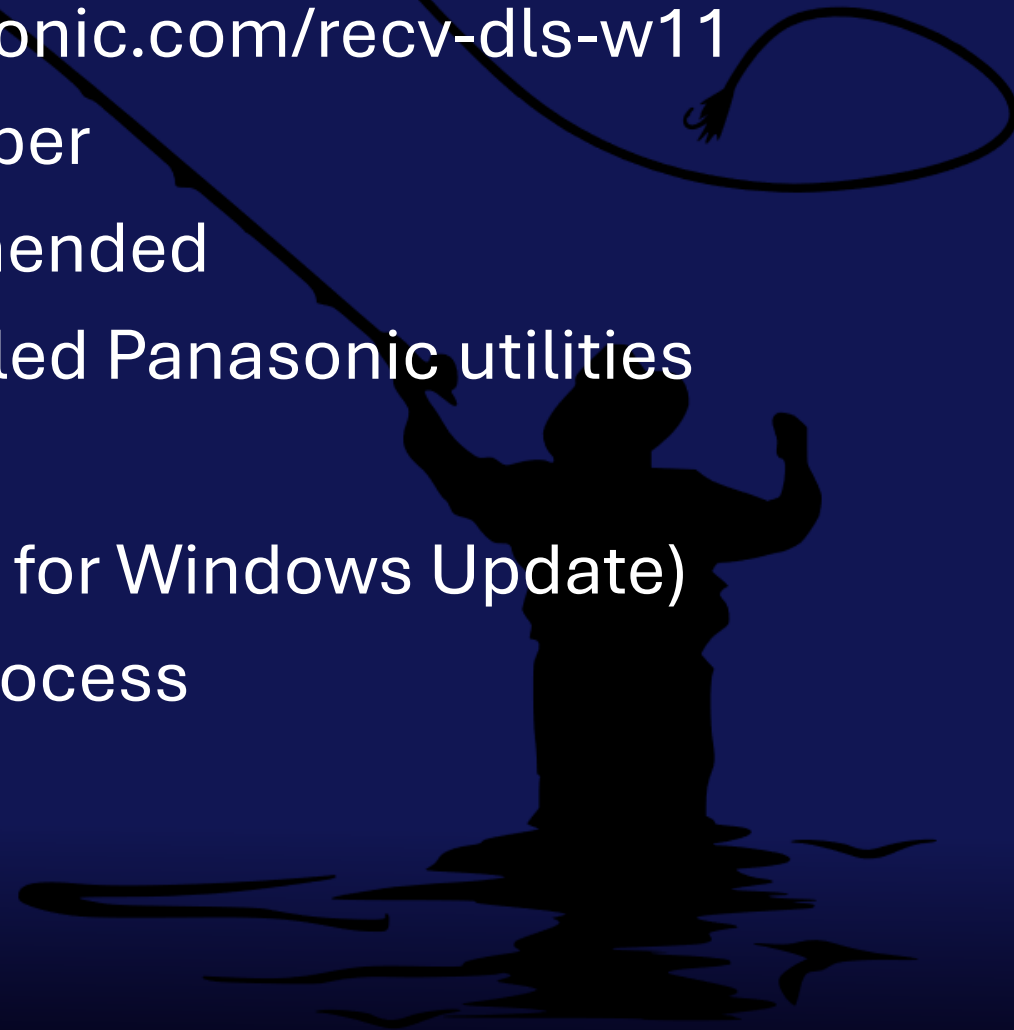
Boot to BIOS setup using F2 > Exit tab > Recovery Partition
(Some modes F8 or F11)

- ◆ Requires intact Panasonic recovery partition
- ◆ Restores factory settings with pre-installed Panasonic utilities
- ◆ Outdated OS/Drivers (longer OOBE time for Windows Update)
- ◆ Input required at different parts of the process
- ◆ Not scalable

Panasonic Recovery Image USB/DVD

<https://global-pc-support.connect.panasonic.com/recv-dls-w11>

- ◆ Requires model number and serial number
- ◆ USB (or DVD) – 16GB or more is recommended
- ◆ Restores factory settings with pre-installed Panasonic utilities
- ◆ 12GB download
- ◆ Outdated OS/Drivers (longer OOBE time for Windows Update)
- ◆ Input required at different parts of the process
- ◆ Not scalable



Surface Windows Recovery Options



Surface Recovery Image – Older Devices

<https://support.microsoft.com/en-us/surface-recovery-image>

- ◆ Requires serial number, USB key (16GB+), and Microsoft account
- ◆ Restores exact factory image with pre-installed drivers
- ◆ Outdated OS/Drivers (longer OOBE time for Windows Update)
- ◆ Only certain Surface devices (Book, Duo, Go, Hub, Laptop, Laptop Go, Laptop SE, Laptop Studio, Pro, Studio, Dev Kit 2023)
- ◆ Not scalable

① Select your Surface

Choose a product and enter the serial number. [Find your serial number](#)

Product

Surface Book



Serial Number *

Required

0182334*****

Continue

Serial Number Location



② Download the Surface recovery image

Important: Don't download the files directly to your USB drive.

Surface recovery images

Download the desired image file and save it to your Surface or another PC. The file will be downloaded as a zip file.

Product	Download link
Surface Book 256GB i5 - Windows 10 Pro Version 20H2	↓ Download image
Surface Book 256GB i5 - Windows 10 Pro Version 2004	↓ Download image

Create the USB recovery drive

If you are on a Surface Hub 2S or Surface Hub 3, see [Reset & recovery for Surface Hub 2S & Surface Hub 3 - Surface Hub](#).

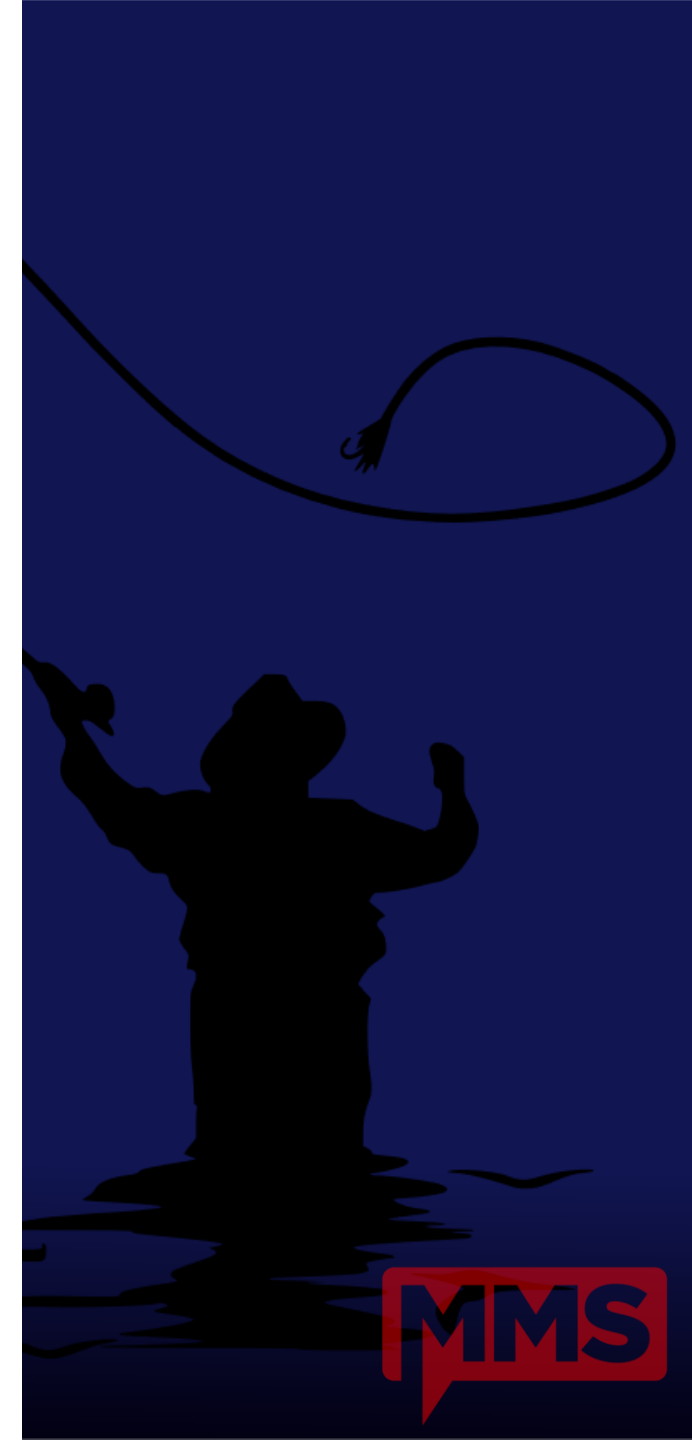
If you are on Surface Duo, see [Recover Surface Duo if it won't start](#).

If you are on other Surface devices, here's how to create the USB recovery drive:

Important: Creating a recovery drive will erase everything that's stored on your USB drive. Make sure you're using a blank USB drive, or be sure to transfer any important data on your USB drive to another storage device before using it to create a recovery drive.

1. Make sure the PC you downloaded the recovery image to is turned on and plugged in, then insert the USB recovery drive into the USB port. (Use a USB 3.0 drive if you can.)
2. In the search box on the taskbar, enter **recovery drive**, then select **Create a recovery drive** or **Recovery Drive** from the results. You might be asked to enter an admin password or confirm your choice.
3. In the **User Account Control** box, select **Yes**.
4. Make sure to clear the **Back up system files to the recovery drive** check box and then select **Next**.
5. Select your USB drive, and then select **Next > Create**. Some utilities need to be copied to the recovery drive, so this might take a few minutes.
6. When the recovery drive is ready, select **Finish**.
7. Double-click the recovery image .zip file that you previously downloaded to open it.
8. Select all the files from the recovery image folder, copy them to the USB recovery drive you created, and then select **Choose to replace the files in the destination**.
9. Once the files have finished copying, select the **Safely Remove Hardware and Eject Media** icon on the taskbar, and remove your USB.

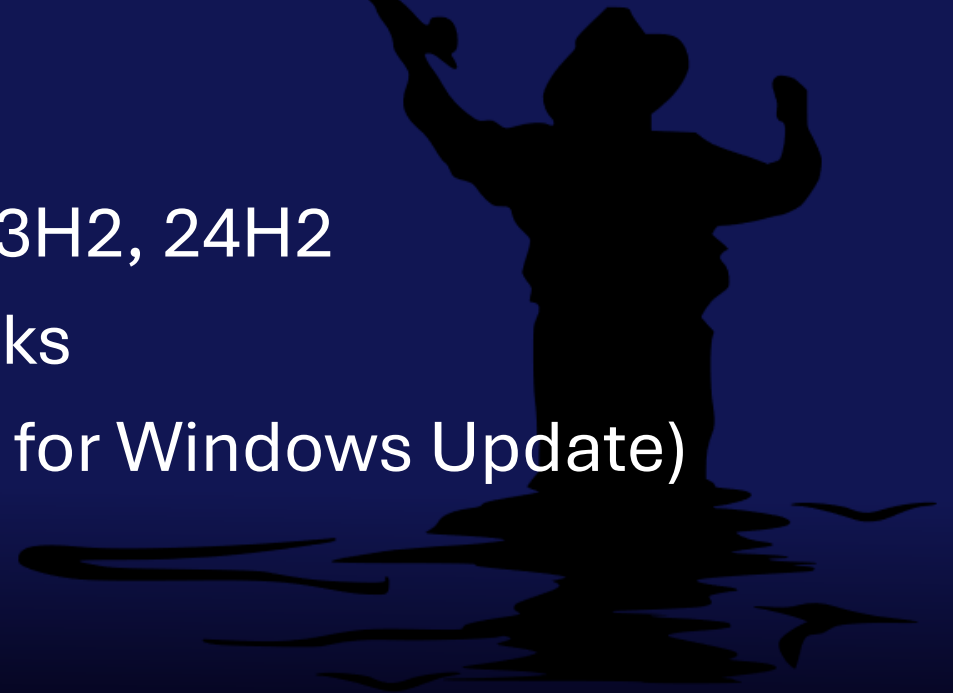
For more info on how to use the image, see [Creating and using a USB recovery drive for Surface](#).



Surface IT Toolkit – Newer Devices

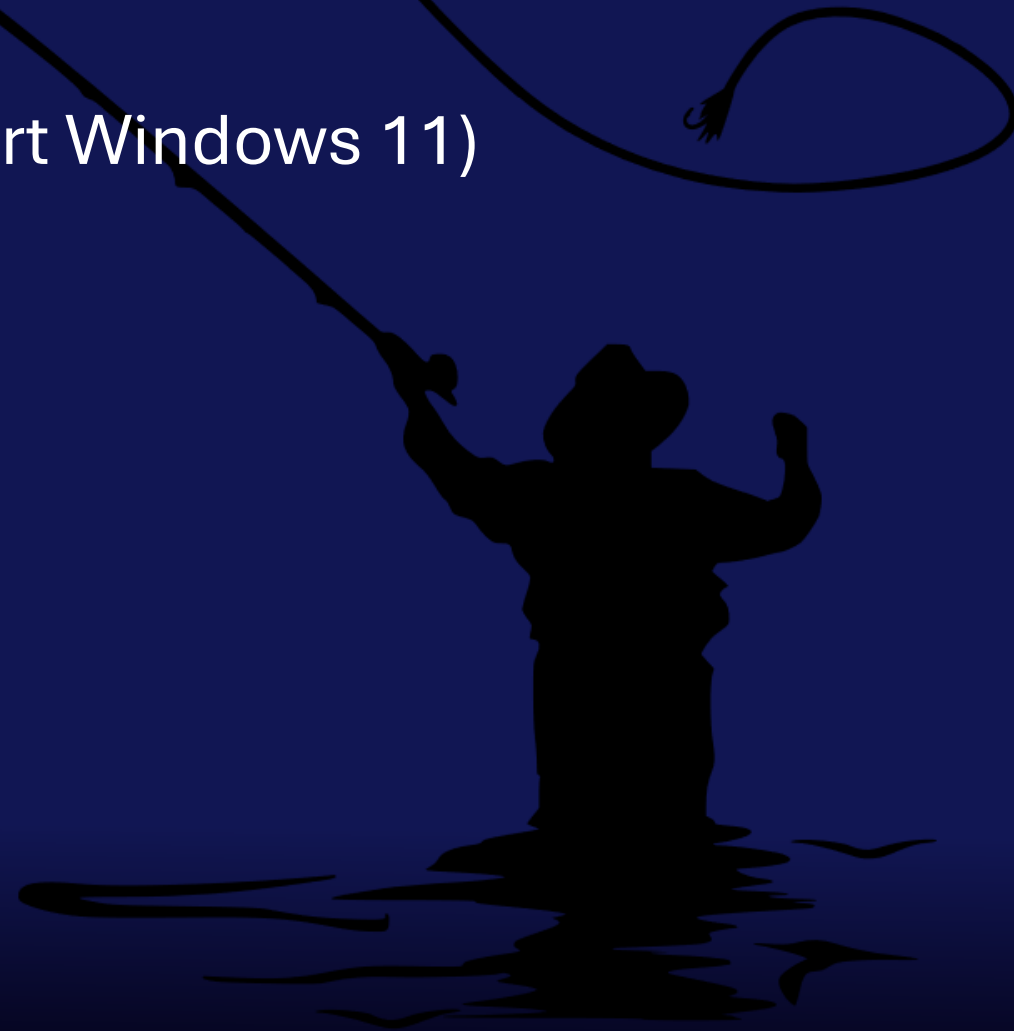
<https://learn.microsoft.com/en-us/surface/surface-it-toolkit>

- ◆ Requires USB key (16GB+)
- ◆ Works on newer Surface devices
- ◆ Ability to select language
- ◆ Supports Windows 10 and Windows 11
- ◆ Includes multiple releases – i.e. 22H2, 23H2, 24H2
- ◆ Option to include multiple language packs
- ◆ Outdated OS/Drivers (longer OOB time for Windows Update)
- ◆ Not scalable



Other Windows Installation Options

- ◆ Configuration Manager OSD
- ◆ MDT (EOL/Deprecated – does not support Windows 11)
- ◆ PSD (Depends on MDT)
- ◆ OSDCloud
- ◆ DeployR/DeployR Community Edition
- ◆ Many others...



A silhouette of a cowboy standing in water, swinging a lasso that forms a large loop in the air. The background is a solid dark blue.

Secure Boot Certificates & Windows Recovery Options

The fun is just beginning...

Secure boot



Selected boot image did not authenticate.
Please press enter to attempt boot to next device.

EliteBook 840 G10 – After over 2 hours of waiting on Wi-Fi to download the
Sure Recover Boot Wim.....

Can you guess why this happened....





One-Time Boot Settings

Control the boot flow for the SupportAssist OS Recovery Tool.

NOTE:

Once a system and/or admin password is set, the system will always prompt for system and/or admin password during boot.

UEFI Boot Devices



Windows Boot Manager



UEFI RST KBG50ZNS256G NVMe
KIOXIA 256GB 33IC13SYEW9X



UEFI Kingston DataTraveler 3.0
A9654DB818C4000000008288



UEFI HTTPs Boot



ONBOARD NIC (IPV4)



ONBOARD NIC (IPV6)

Pre-Boot Tasks

Change important BIOS settings on your system, configure how your device works and troubleshoot issues using this interface.

BIOS SETUP

Configure BIOS options and control how your system functions.

DIAGNOSTICS

Run system tests to identify any issues.

BIOS UPDATE

Search for and install the latest BIOS updates from various services.

SupportAssist OS Recovery

Analyze, repair and restore your system.

BIOS Flash Update - Remote

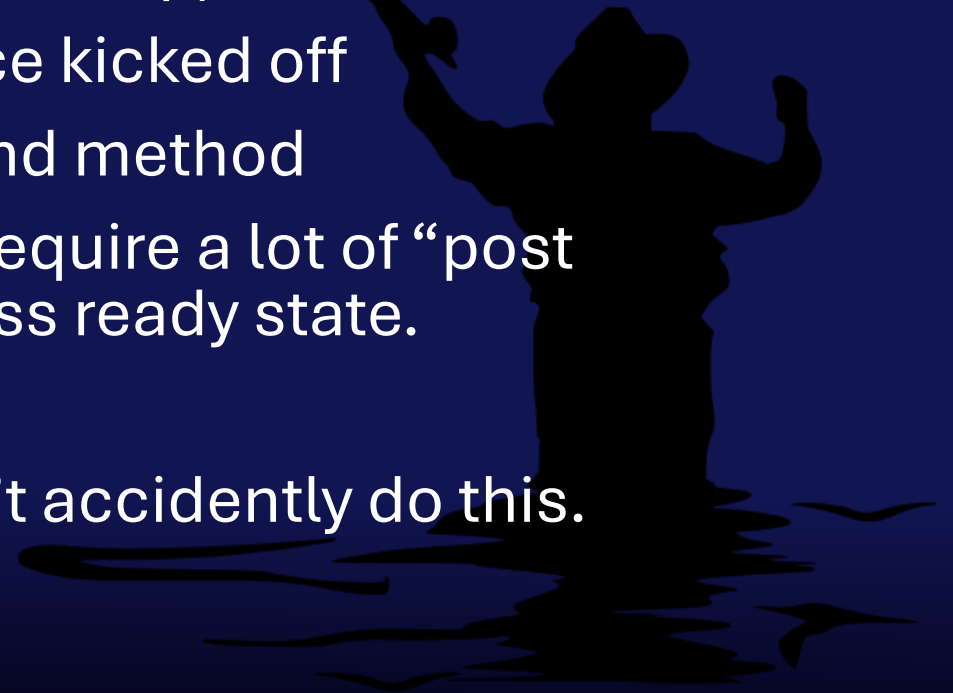
BIOS and Firmware Update Over-the-Air

Device Configuration

Configure device settings

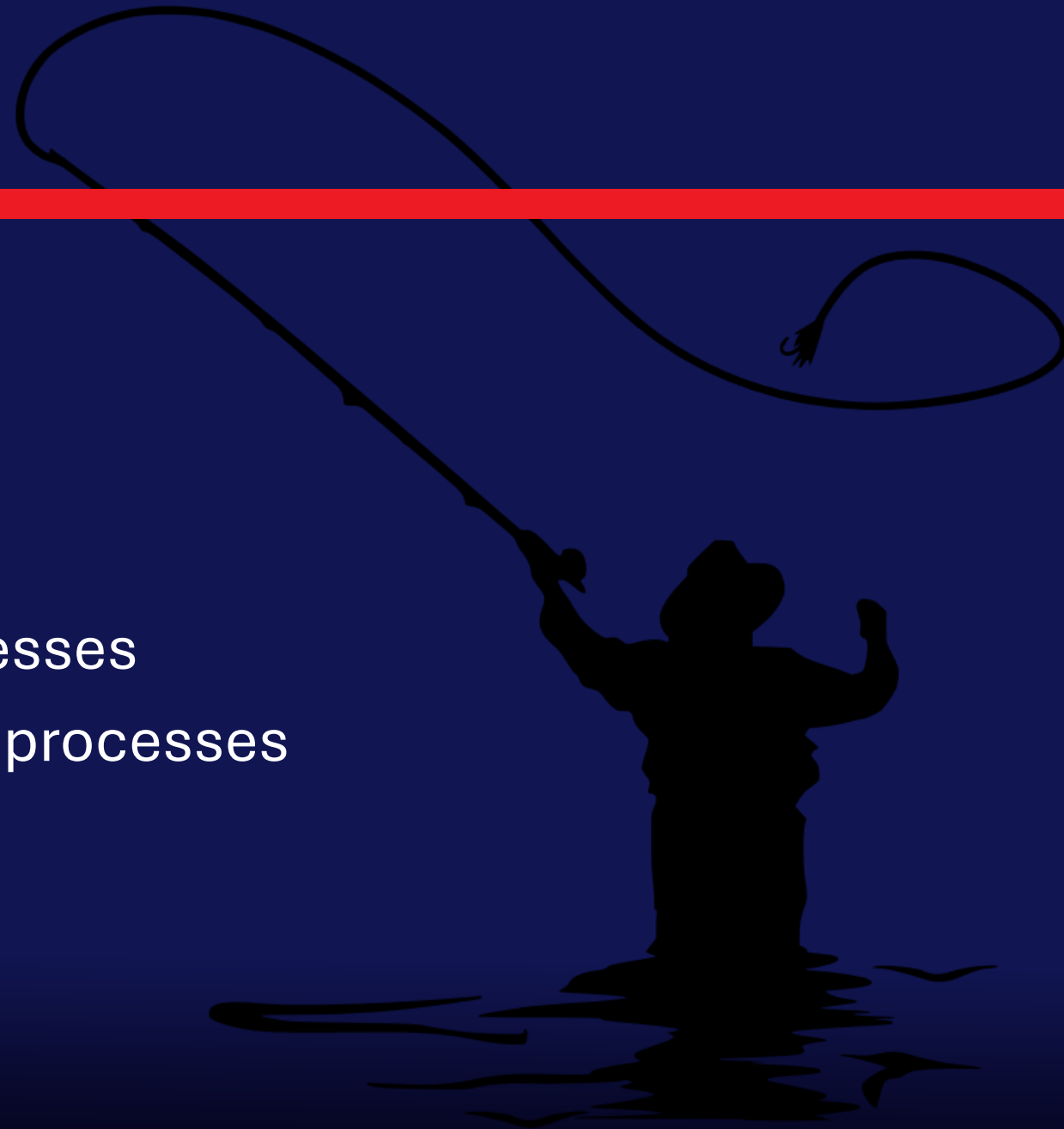
OEM recovery lessons learned

- ◆ Keep your BIOS/UEFI updated!
- ◆ Wi-Fi... hit and miss, very inconsistent
- ◆ Not all USB Adapters created equal
- ◆ Not all Flash Drives created equal (don't go cheap)
- ◆ Not all OEMs processes are Zero Touch once kicked off
- ◆ Wide variety of processes based on OEM and method
- ◆ End results are also quite different, would require a lot of “post recovery management” to get into a business ready state.
- ◆ Not scalable, not for a novice end user
- ◆ Consider: Locking down so end users don't accidentally do this.



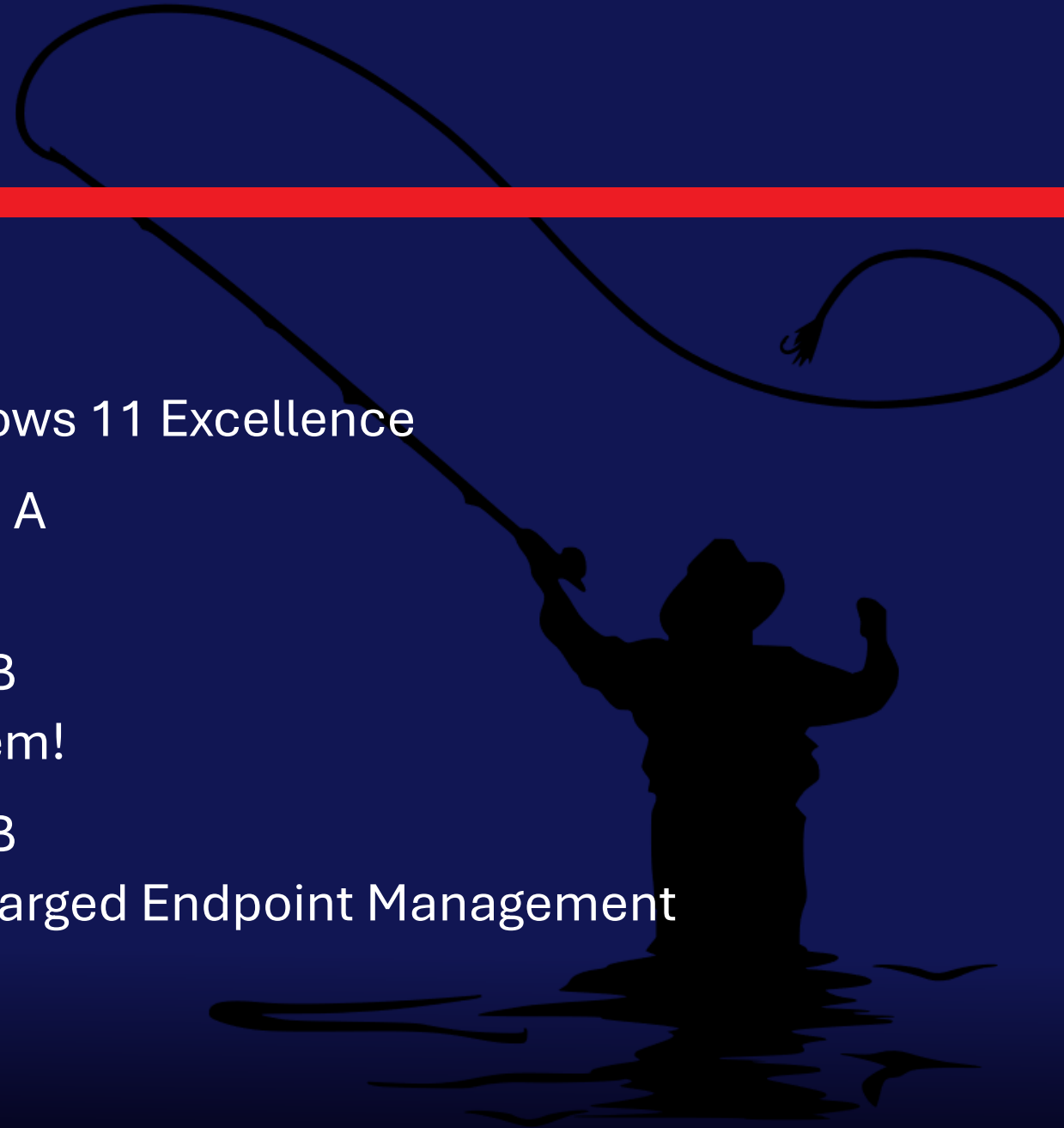
Start Today

- ◆ Develop a plan of action
 - ◆ Classify critical endpoints
 - ◆ Alternate Windows OS (W365/VDI)
 - ◆ Repair Windows OS
 - ◆ Re-install Windows OS
- ◆ Develop Windows recovery processes
- ◆ Routinely test Windows recovery processes



Other Sessions

- ◆ OSD Essentials for the Modern Admin
Mike Terrill & Gary Blok
Monday, May 4th at 3:00 PM Lakes B
- ◆ Beyond Deployment: Sustaining Windows 11 Excellence
Johan Arwidmark & Mike Terrill
Wednesday, May 6th at 10:00 AM Lakes A
- ◆ Secure Boot Certificates: 2026 Edition
Mike Terrill & Gary Blok
Wednesday, May 6th at 1:00 PM Lakes B
- ◆ Build Your Test Lab: No Lab? No Problem!
Mattias Benninge & Mike Terrill
Wednesday, May 6th at 5:00 PM Lakes B
- ◆ PowerShell & GitHub Copilot: Supercharged Endpoint Management
Gary Blok & Carl Johan Hederöth
Thursday, May 7th at 10:00 AM Lakes B



References

<https://zerodayclock.com/>

<https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>

https://www.eiopa.europa.eu/digital-operational-resilience-act-dora_en

<https://digital-strategy.ec.europa.eu/en/policies/nis2-directive>

<https://www.sec.gov/rules-regulations/2023/07/s7-09-22>

<https://www.hhs.gov/hipaa/index.html>



References

<https://www.microsoft.com/en-us/software-download/windows11>

<https://www.dell.com/support/contents/en-do/article/product-support/self-support-knowledgebase/software-and-downloads/dell-os-recovery-tool>

https://support.hp.com/us-en/document/ish_4511095-4511141-16

<https://pcsupport.lenovo.com/us/en/lenovorecovery>

<https://global-pc-support.connect.panasonic.com/recv-dls-w11>

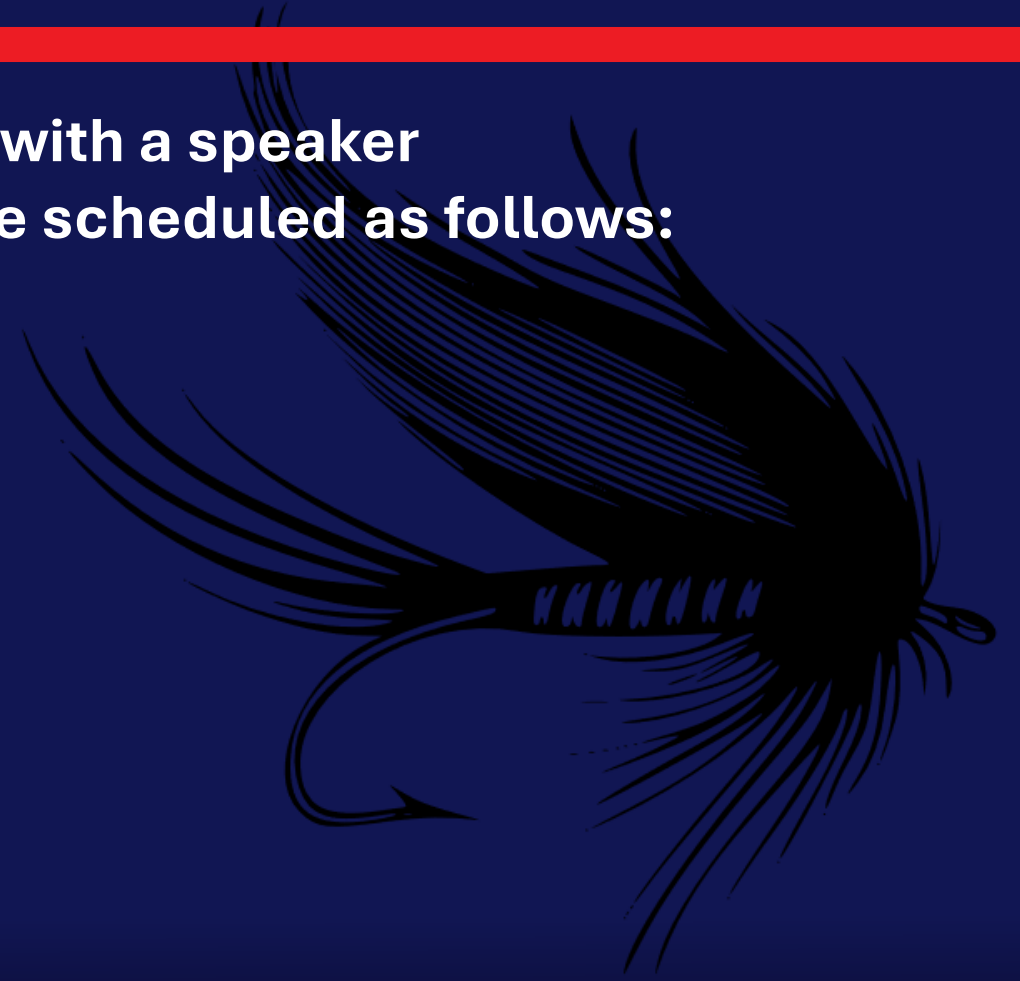
<https://support.microsoft.com/en-us/surface-recovery-image>

<https://learn.microsoft.com/en-us/surface/surface-it-toolkit>

Fishing Sessions

**Be sure to find Wally and sign up for 1:1 time with a speaker
If you'd like to "catch" us for a session, we are scheduled as follows:**

- ◆ Mike: 5/5/2026 8:00 AM
- ◆ Mike: 5/5/2026 8:20 AM
- ◆ Mike: 5/5/2026 8:40 AM
- ◆ Mike: 5/5/2026 9:00 AM
- ◆ Mike: 5/5/2026 9:20 AM



Don't forget to leave feedback for your speakers!

SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



ninjaOne®



numecent®

